

HOW PRETRAINED SPECTRAL SUBSPACES AND THEIR INTERACTIONS AFFECT FINE-TUNING

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ABSTRACT

Many fine-tuning methods use the singular value decomposition of pretrained weights to choose which directions to adapt. Can a model learn when its updates stay entirely in the lower spectral tail, the directions with smaller singular values? Does allowing those directions to interact improve adaptation? We introduce a new family of adapters that separates three choices: changing individual tail coefficients, combining directions within the tail, and allowing interaction with leading directions. Our analysis explains the freedom each choice provides and how to control it. In RoBERTa experiments on RTE, training only tail coefficients raises accuracy from 60% with a trained classifier alone to 70%. Leading and middle directions reach similar accuracy, and rotations give no consistent further gain. In a separate experiment allowing both leading and tail updates, restricting their interaction roughly halves held-out prediction loss on RTE and MRPC compared with a size penalty tuned for similar update sizes, while mean accuracy changes by less than half a percentage point. In these experiments, accuracy is similar across subspace choices and interaction settings, while interaction control can improve the probabilities assigned to answers.

1 INTRODUCTION

Fine-tuning adapts a pretrained model to a new task. A growing family of methods uses the singular value decomposition (SVD) of pretrained weights to choose which directions to train. Some begin with the leading directions, associated with larger singular values. Others begin with the lower spectral tail, associated with smaller values, or learn coefficients and rotations in a fixed singular-vector basis (Meng et al., 2024; Wang et al., 2025b; Han et al., 2023; Lingam et al., 2024; Zhang & Pilanci, 2024; Wu et al., 2026). These approaches make the pretrained spectrum part of the adaptation design.

Choosing directions is one part of that design. Another is deciding how they may interact during training. Can the lower tail alone support adaptation? Does combining its directions improve learning? What changes when leading and tail directions can interact? We study these questions by separating where an update is allowed from how freely it can change that region, and measuring the resulting prediction loss and accuracy.

We introduce a family of adapters that separates these choices. LinLoRA learns one coefficient per selected tail direction, changing its singular value. OrthoLoRA adds rotations that shuffle different directions within the tail. Both leave the remaining pretrained directions unchanged. Our analysis quantifies which additional weight changes rotations can express.

Their scaled versions, UILinLoRA and UIOrthoLoRA, also learn diagonal matrices for each input and output coordinate of an adapted layer. These diagonal scalars can shuffle tail and leading directions. We derive an interaction penalty that discourages this shuffling while allowing changes within either group.

We run two studies with RoBERTa-base. The first tests tail adaptation on RTE (sentence entailment). We attach a small classifier after RoBERTa’s final layer to map its sentence-pair representation to label probabilities. We train this classifier either alone or together with LinLoRA or OrthoLoRA adapters in all 48 attention projections. Adapters use 256 tail directions; the pretrained weights remain frozen.

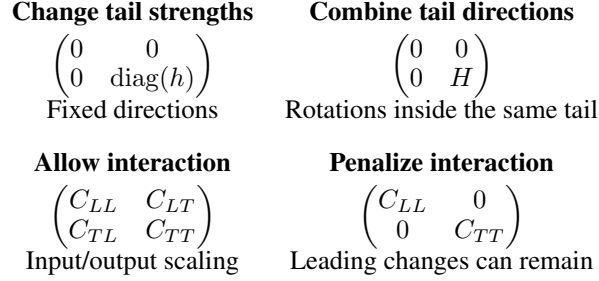


Figure 1: Adapter structure in the original pretrained basis. Top: coefficients and rotations both keep updates inside the tail. Bottom: the scaled adapter permits changes in both groups and between them; its zero-interaction limit retains the two within-group blocks. The diagrams show the update patterns allowed by each construction.

The tail supports useful adaptation beyond the classifier alone while leading directions stay fixed. Leading and middle directions also support adaptation, showing that useful learning is possible in each tested spectral region. Rotations add flexibility but give no consistent further accuracy gain (Table 1). Section 5.1 describes the training and evaluation design.

The second study asks whether controlling interaction offers more than shrinking weight changes. On RTE and MRPC (paraphrase detection), we train UILinLoRA with both leading and tail updates and the same classifier design. We compare no penalty, our interaction penalty, and a size penalty tuned to give similar update sizes. Interaction control gives lower held-out prediction loss than either alternative, while mean accuracy changes little (Table 2 and Figure 2). The gain lies in the probabilities assigned to answers and would be missed by looking only at accuracy. Section 5.2 gives the training and evaluation design.

The penalties also produce different update patterns. Interaction control reduces the share of weight change connecting leading and tail directions; shrinking updates leaves that share high (Figure 3). Update size alone therefore misses how adaptation is distributed across directions. This loss improvement is achieved while both leading and tail directions are allowed to change.

2 A FAMILY OF CONTROLLED SPECTRAL ADAPTERS

2.1 START WITH A FIXED SPECTRAL TAIL

For a square pretrained weight matrix $W_0 \in \mathbb{R}^{d \times d}$, write

$$W_0 = U \Sigma V^\top = U_L \Sigma_L V_L^\top + U_T \Sigma_T V_T^\top. \quad (1)$$

Singular values are ordered from largest to smallest. We call the first k directions leading (L), and the remaining $r = d - k$ directions the tail (T). The cutoff divides the pretrained directions into two groups; both can have nonzero singular values. We keep these original bases fixed throughout training and analysis of the confined adapters. Appendix A.4 extends the construction to rectangular weights.

An adapted layer is $W_{\text{eff}} = W_0 + \Delta$. To guarantee that its update stays in the selected tail, use

$$\Delta = U_T H V_T^\top. \quad (2)$$

Only the small matrix H is learned. Multiplication by the fixed tail bases keeps every update inside the selected region. We can therefore change the freedom of H while holding the allowed region constant.

2.2 CHANGE COEFFICIENTS, THEN ALLOW ROTATIONS

The first adapter sets $H = \text{diag}(h)$, with $h \in \mathbb{R}^r$ trainable. It changes the strength of each selected pair of pretrained directions independently. We use LinLoRA as its implementation label. SVDiff and SVFT already establish fixed-basis coefficient tuning (Han et al., 2023; Lingam et al., 2024);

SVFT also learns selected off-diagonal coefficients. Here the diagonal family provides a simple restricted reference.

The second adapter allows two learned orthogonal rotations inside the last $q \leq r$ tail directions. Split the tail into $r - q$ fixed directions (M) and q rotatable directions (S), and set

$$H = \begin{bmatrix} \text{diag}(h_M) & 0 \\ 0 & R_U \text{diag}(h_S) R_V^\top \end{bmatrix}, \quad R_U, R_V \in \mathbb{O}(q). \quad (3)$$

This is the OrthoLoRA construction. At $q = 0$, it reduces to fixed coefficients. At $q = r$, arbitrary coefficients and the inner SVD can represent any real tail matrix H . Intermediate values give more freedom to a smaller block. The implementation settings $k_{\text{val}} = r$ and $k_{\text{vec}} = q$ specify these two dimensions. The rotation size q controls the added freedom and the number of trainable parameters. Our controlled study enforces orthogonality through a matrix exponential (Appendix B.19).

Both adapters remain inside the same tail. The difference is whether a tail input direction can connect only to its paired output direction or also to other output directions in the tail. This isolates flexibility within the selected region.

2.3 SEPARATE LOCATION FROM FLEXIBILITY

The following result quantifies the extra freedom supplied by a full tail core. Let G be a target weight update; $\|\cdot\|_F$ denotes the Frobenius norm, the square root of the sum of squared entries.

Proposition 1 (Two distinct approximation bottlenecks). *For an arbitrary target matrix $G \in \mathbb{R}^{d \times d}$, let $G_{TT} = U_T^\top G V_T$. Then*

$$\min_h \|G - U_T \text{diag}(h) V_T^\top\|_F^2 = \|G\|_F^2 - \sum_{i=1}^r |(G_{TT})_{ii}|^2, \quad (4)$$

$$\min_H \|G - U_T H V_T^\top\|_F^2 = \|G\|_F^2 - \|G_{TT}\|_F^2. \quad (5)$$

The improvement in the best attainable squared error equals the off-diagonal energy of G_{TT} . Both families retain the same unavoidable outside-tail residual.

For example, connecting one tail input direction to a different tail output direction requires an off-diagonal entry of H . A full tail core can express that connection. Diagonal coefficients leave it out. Proposition 1 separates this within-tail restriction from the residual outside the selected region. The proof follows by orthogonal projection (Appendix A.1). Training determines which of the permitted updates is learned.

Confinement is defined relative to the original layer coordinates. For the same input x , $U_L^\top (W_0 + \Delta)x = U_L^\top W_0 x$. The network still learns through the changed tail outputs and through inputs altered by preceding layers. The original leading directions remain top-ranked when

$$\sigma_k(W_0) > \|\Sigma_T + H\|_2. \quad (6)$$

Appendix A.3 gives a counterexample when this separation fails.

3 CONTROLLING LEADING-TAIL INTERACTION

The confined families test what can be learned inside the tail. To study a relaxation, add diagonal output and input scalars E, D . They act in the layer’s original coordinates, rather than in its singular-vector basis. Consequently, $EU_T H V_T^\top D$ can affect directions outside the tail.

The scaled adapter used in our main experiment also includes a leading core. We write both choices in one form:

$$\Delta = E(U_L S_L V_L^\top + U_T H V_T^\top) D, \quad S_L = \begin{cases} 0 & \text{tail core only,} \\ I & \text{practical adapter.} \end{cases} \quad (7)$$

The expression is added to W_0 . Thus $S_L = I$ adds an identity core in the leading coordinates, and learning E, D changes its contribution along with the tail contribution. We call this the practical adapter. UILinLoRA uses diagonal H ; UIOrthoLoRA also includes the rotations. Our completed interaction experiment uses the practical UILinLoRA form.

3.1 IDENTIFY THE INTERACTIONS TO CONTROL

We inspect the effective update in the original pretrained basis:

$$C = U^\top \Delta V = \begin{bmatrix} C_{LL} & C_{LT} \\ C_{TL} & C_{TT} \end{bmatrix}. \quad (8)$$

C_{TT} records tail-to-tail changes, including any within-tail combinations. C_{LL} records changes among leading directions. The two cross blocks link the leading and tail groups. Strict tail confinement leaves only C_{TT} . Decoupling the two groups instead leaves both within-group blocks, C_{LL} and C_{TT} .

Let $A = U^\top EU$, $B = V^\top DV$, and define the two scaler-mixing operators

$$M_E = U_L^\top EU_T, \quad M_D = V_L^\top DV_T, \quad \mu = \|M_E\|_2, \quad \nu = \|M_D\|_2. \quad (9)$$

The norm $\|\cdot\|_2$ is the operator norm. Direct multiplication gives

$$C_{ab} = A_{aL}S_LB_{Lb} + A_{aT}HB_{Tb}, \quad a, b \in \{L, T\}. \quad (10)$$

The two operators measure how the scalars connect the groups. When both are zero, the cross blocks vanish. A tail-only core then recovers strict confinement. The practical core retains its leading block: for example, $E = D = I, H = 0$ gives $C_{LL} = I$. With bounded factor norms, small mixing operators bound the cross blocks as stated in Proposition 2 and Corollary 3 (Appendix A.2).

3.2 HOW THE INTERACTION PENALTY WORKS

We train with

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \sum_{\ell \in \mathcal{M}} (\lambda_E \|M_{E,\ell}\|_F^2 + \lambda_D \|M_{D,\ell}\|_F^2), \quad (11)$$

where $\mathcal{M}$ contains the adapted matrices. The two coefficients control the pressure on output-side and input-side interaction. Task training and the interaction penalty jointly determine the update.

The penalty’s mechanism is explicit. Write $P = U_L U_L^\top$ and $E = \text{diag}(\epsilon)$. Then

$$\|M_E\|_F^2 = \frac{1}{2} \|EP - PE\|_F^2 = \sum_{i < j} P_{ij}^2 (\epsilon_i - \epsilon_j)^2. \quad (12)$$

It penalizes differences between scaling factors, with weights determined by the pretrained projector. If the associated graph is connected, its zero-penalty solutions are uniform scalars. Isolated penalty dynamics then approach $E = aI, D = bI$. For the practical core this gives $C_{LL} = abI$, $C_{TT} = abH$, and zero cross blocks. This gives the mechanism of the penalty: it suppresses connections between the groups while retaining a uniformly scaled leading contribution. During adaptation, task gradients act alongside this pressure. Our experiment measures the resulting interaction, loss and accuracy.

We compare this intervention with penalizing the size of the effective update. The two objectives act on different quantities: one penalizes scaler interactions, while the other penalizes total displacement from the pretrained weights. Their behavior during factor training is analyzed in Appendices A.7–A.10. We evaluate the final effective update directly, which also accounts for reciprocal rescalings of the adapter factors.

4 MEASURING ADAPTATION

We evaluate behavior and weight changes at the same saved checkpoint. Behavior tells us how well the adapted model predicts. The weight measurements show what the adapter changed to produce those predictions.

Prediction loss and accuracy. Accuracy counts correct predicted labels. Prediction loss is cross-entropy, the mean negative log probability assigned to the correct label. Two models can predict the same label while assigning it different probabilities, so loss can reveal a difference that accuracy misses. We report both on held-out task examples at the fixed training endpoint. The three paired training seeds are the units of replication; tables give means and sample standard deviations, with paired differences in the appendix.

Table 1: RTE adaptation in leading, middle and tail directions. Accuracy and cross-entropy (lower is better) are fixed-endpoint inner-selection results, mean $\pm$ sample SD over three paired seeds. Every condition trains the same 592,130-parameter classifier; adapter counts exclude it.

Trainable backbone update	Accuracy (%)	Task loss (nat)	Adapter params
Classifier only	59.77 $\pm$ 0.65	0.654 $\pm$ 0.002	0
Leading coefficients	71.75 $\pm$ 1.71	1.785 $\pm$ 0.083	12,288
Leading coefficients + rotations	72.22 $\pm$ 1.01	2.075 $\pm$ 0.118	405,504
Middle coefficients	71.02 $\pm$ 1.21	1.759 $\pm$ 0.027	12,288
Middle coefficients + rotations	72.22 $\pm$ 1.21	1.958 $\pm$ 0.044	405,504
Tail coefficients	70.01 $\pm$ 0.23	2.001 $\pm$ 0.141	12,288
Tail coefficients + rotations	71.02 $\pm$ 3.30	2.111 $\pm$ 0.268	405,504

Interaction between leading and tail directions. The four blocks in (8) partition the squared update norm:

$$\|\Delta\|_F^2 = \sum_{a,b \in \{L,T\}} \|C_{ab}\|_F^2. \quad (13)$$

For a nonzero update, $p_{ab} = \|C_{ab}\|_F^2 / \|\Delta\|_F^2$ is the fraction of energy in a block. We call $p_{\text{cross}} = p_{LT} + p_{TL}$ the cross share: it measures the share connecting leading and tail directions. The total share outside the tail is $p_{\text{off}} = p_{LL} + p_{\text{cross}}$. Off-diagonal entries *inside* C_{TT} instead describe combinations within the tail. These measurements distinguish the three choices in our adapter family. Fractions at an exactly zero update are undefined.

Overall size and distribution across the network. Relative update size is $\rho_F = \|\Delta\|_F / \|W_0\|_F$. We use it to compare interaction control with reducing the overall amount of weight change. For the network-wide summary, we sum squared energies across adapted matrices before taking ratios. We also report each matrix’s norm, the equal-module mean of block fractions, and the cumulative energy carried by the largest updates. These views distinguish a broadly distributed change from one concentrated in a few matrices. Appendix B gives the exact aggregations, including the archived ratio $q_{\text{off}} = \sqrt{p_{\text{off}}}$.

Changes learned after insertion. The practical adapter starts with a nonzero contribution. We therefore measure both its total displacement from the pretrained weights, $\Delta = W_{\text{eff}} - W_0$, and its displacement during training, $\Delta_{\text{learned}} = W_{\text{eff}} - W_{\text{eff}}(0)$. We subtract matrices before computing energies. Within the learned leading block, we also separate its scalar identity component from the remaining structure. This connects the measured endpoint to the uniform-scaler limit derived in Section 3.

A separate masked-token behavior. We evaluate a fixed masked-token probe using the original frozen language-model head. It measures a pretraining-style behavior on the same adapted checkpoints. The task head and language-model head test different predictions; their losses are reported separately.

5 EXPERIMENTS: TAIL ADAPTATION AND INTERACTION CONTROL

5.1 THE TAIL SUPPORTS ADAPTATION BEYOND THE TRAINED HEAD

On RTE, we compare classifier-only training on frozen RoBERTa-base with six adapters: coefficients alone or coefficients plus rotations, using 256 leading, middle or tail directions in all 48 attention projections. Every condition trains the same classification head, which maps sentence-pair representations to two label probabilities. Pretrained weights remain frozen; adapters start at zero and, when enabled, rotate 64 directions. Conditions share head initialization, data order and 5,670 optimizer steps within each of three seeds. We train on 80% of RTE’s training set and report accuracy and cross-entropy at the final step on the remaining 20% (inner-selection). The official validation set remains unevaluated for this study. Appendix B.19 gives optimizer settings and individual runs.

Table 2: Interaction control, size control and unregularized training at the same fixed endpoints. All conditions use the practical leading-plus-tail adapter without rotations. Entries are mean $\pm$ sample SD over three seeds. Accuracy and prediction loss use held-out task examples; Cross is the pooled percentage of update energy connecting leading and tail directions, and ρ_F is relative update size. Table 8 gives per-seed size differences after calibration.

Task	Condition	Accuracy (%)	Task loss (nat)	Cross (%)	ρ_F
RTE	UNREG	71.72 $\pm$ 1.50	2.429 $\pm$ 0.071	43.7 $\pm$ 0.0	0.2751 $\pm$ 0.0041
RTE	MIX	71.36 $\pm$ 1.63	1.229 $\pm$ 0.089	10.0 $\pm$ 0.5	0.0736 $\pm$ 0.0046
RTE	NORM	71.24 $\pm$ 1.63	2.193 $\pm$ 0.327	46.2 $\pm$ 0.0	0.0759 $\pm$ 0.0016
MRPC	UNREG	86.19 $\pm$ 0.93	0.871 $\pm$ 0.068	42.4 $\pm$ 0.1	0.1802 $\pm$ 0.0009
MRPC	MIX	85.46 $\pm$ 1.02	0.395 $\pm$ 0.028	4.5 $\pm$ 0.3	0.0407 $\pm$ 0.0017
MRPC	NORM	85.78 $\pm$ 0.65	0.755 $\pm$ 0.064	46.2 $\pm$ 0.1	0.0423 $\pm$ 0.0005

Diagonal coefficients improve accuracy over classifier-only training in each tested spectral region (Table 1). Useful adaptation is therefore possible in several parts of the spectrum. The band comparisons do not establish equal performance across regions (Table 23). Rotations expand the allowed updates without a consistent accuracy gain. Every adapter has higher cross-entropy than the classifier alone, and rotations increase mean loss in every region. Accuracy gains and added flexibility therefore do not guarantee better label probabilities.

5.2 EXPERIMENTAL DESIGN FOR INTERACTION CONTROL

The main experiment uses RoBERTa-base (Liu et al., 2019) on RTE, a textual-entailment task, and MRPC, a paraphrase task. We freeze the backbone and train adapters in all 48 attention projections together with the task head. Each 768×768 projection has 256 tail coefficients and input/output scalers, with the practical leading core $S_L = I$ and no rotations. Thus each condition can change both leading and tail directions; the intervention controls how those groups interact.

We compare three conditions. UNREG trains only on the task loss. MIX adds the two-sided interaction penalty with $\lambda_E = \lambda_D = 10^{-3}$. NORM adds the size penalty $\beta \text{mean}_\ell \|\Delta_\ell\|_F^2 / \|W_{0,\ell}\|_F^2$. The size control asks whether the effect of MIX is reproduced by making the overall update smaller. Its dose is selected using update size on a separate calibration seed, without using task scores.

Each condition uses confirmation seeds 17, 42 and 123, giving 18 runs across the two tasks. Conditions share initialization, data order, optimizer settings and step budgets within a task and seed. Training ends at 5,670 steps on RTE and 2,760 on MRPC. After freezing these endpoints, we evaluate once on the previously held-out official validation sets: 277 examples for RTE and 408 for MRPC. All runs pass checkpoint reload and metric-reproduction checks. Appendix B.8 records the complete calibration frontier, per-seed matches and inner-selection outcomes; Appendix B.12 gives the executed recipes.

5.3 INTERACTION CONTROL LOWERS PREDICTION LOSS

MIX has lower held-out prediction loss than both controls in every paired seed (Table 2).

Relative to NORM, mean loss falls from 2.193 to 1.229 on RTE and from 0.755 to 0.395 on MRPC: reductions of 44% and 48%. Mean accuracy is 71.36% versus 71.24% on RTE and 85.46% versus 85.78% on MRPC for MIX and NORM. Their differences are +0.12 and -0.33 percentage points. Figure 2 shows every seed: the loss values separate by condition much more clearly than the accuracies. The result concerns the probabilities assigned to answers, which an accuracy-only comparison would miss. Section 7 discusses confidence calibration and the precision of the accuracy comparison.

5.4 SMALLER UPDATES RETAIN SUBSTANTIAL INTERACTION

The two penalties produce different update patterns. At its selected doses, NORM retains 46.2% cross share on both tasks. MIX has $10.0 \pm 0.5\%$ on RTE and $4.5 \pm 0.3\%$ on MRPC, compared with

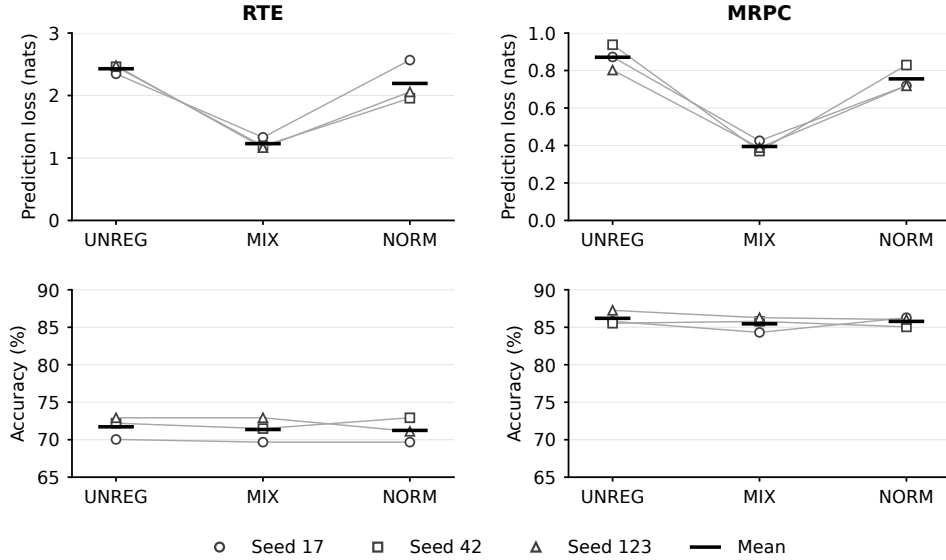


Figure 2: Held-out prediction loss and accuracy at the fixed endpoints. Symbols identify seeds 17, 42 and 123; gray lines connect the same seed across conditions, and black bars mark means. MIX has lower loss than both alternatives in every pair. Table 2 gives the means and sample standard deviations.

the unregularized 43.7% and 42.4%. Interaction control therefore changes the relative allocation of the update as well as its size.

The calibration procedure selects $\beta = 7.5$ for RTE and 27 for MRPC. Their overall sizes differ from the MIX targets by 4.1% and 2.3% on the calibration seed. Across confirmation seeds, absolute differences range from 0.1 to 8.0%; Table 8 gives each signed error. We retain the frozen doses and all confirmation pairs.

The full seven-dose sweep per task provides a broader view (Figure 3). On the calibration seed, NORM reduces update size by $12.1\times$ on RTE and $8.5\times$ on MRPC. Cross share remains 43.7–50.9% and 42.4–48.6%, respectively. None of these tested doses reproduces MIX’s low cross share. The distinction is visible across a wide range of achieved sizes, as well as at the selected doses.

5.5 INTERACTION CONTROL REDISTRIBUTES ADAPTATION

The global norm summarizes how much changed, but not where the change occurred. On RTE, the three largest MIX updates carry 61.9–68.3% of total energy, versus 15.2–16.7% for NORM. Between 12 and 15 of MIX’s 48 module-relative norms are below 10^{-6} . On MRPC, its top three modules carry 35.7–50.1%, versus 17.5–19.5% for NORM. Interaction control produces a more concentrated distribution in both tasks. Appendix B.10 shows the individual matrices and complete cumulative curves.

Leading changes remain after subtracting the insertion state: they account for 45.2% and 65.5% of MIX’s pooled learned energy on RTE and MRPC. Of that leading energy, 84.7% and 89.5% lies in module-wise scalar identity components. This is the structure predicted by the uniform-scaler limit. The remaining non-identity leading energy is about 6.9% of total learned energy on both tasks (Appendix B.13). Thus interaction suppression coexists with substantial, mostly identity-shaped leading changes.

Equal-module averages and the small-update sensitivity analysis also retain the cross-share gap (Appendix B.14). Similar global sizes can conceal different module distributions and block allocations.

Table 3: Historical RoBERTa results on six GLUE tasks: reported central values $\pm$ SD, multiplied by 100, with source rounding retained. Imported reference runs (Albert et al., 2025) and spectral runs use separate tuning protocols. MRPC is accuracy throughout. Params are source-reported for imported runs; $\dagger$ marks adapter-only counts derived from the stated spectral configurations rather than verified checkpoint inventories. Appendix C accounts for frozen storage and computation.

Method	SST-2	MRPC Acc.	CoLA	QNLI	RTE	STS-B	Params
RoBERTa-base							
<i>Imported reference runs (RandLoRA protocol; five runs)</i>							
VeRA-1024	91.9 $\pm$ 0.4	88.4 $\pm$ 1.2	59.9 $\pm$ 2.2	90.5 $\pm$ 0.4	74.9 $\pm$ 1.5	90.4 $\pm$ 0.2	0.26M
LoRA-4	94.4 $\pm$ 0.5	87.3 $\pm$ 0.2	58.4 $\pm$ 0.8	92.7 $\pm$ 0.2	71.5 $\pm$ 1.2	90.5 $\pm$ 0.1	0.7M
RandLoRA-64	92.2 $\pm$ 0.3	88.0 $\pm$ 1.5	59.4 $\pm$ 2.1	91.3 $\pm$ 0.4	74.7 $\pm$ 1.9	90.3 $\pm$ 0.2	0.7M
<i>Spectral instruments (reported protocol; six seeds)</i>							
UIOrthoLoRA	94.8 $\pm$ 0.3	92.0 $\pm$ 1.0	64.4 $\pm$ 1.0	92.4 $\pm$ 0.1	78.7 $\pm$ 1.0	90.7 $\pm$ 0.1	0.479M †
UILinLoRA	93.1 $\pm$ 1.0	90.0 $\pm$ 1.0	65.8 $\pm$ 1.0	92.4 $\pm$ 0.2	76.2 $\pm$ 1.0	90.6 $\pm$ 0.1	0.086M †
RoBERTa-large							
<i>Imported reference runs (RandLoRA protocol; five runs)</i>							
VeRA-256	95.8 $\pm$ 0.3	89.3 $\pm$ 1.2	65.3 $\pm$ 1.1	94.1 $\pm$ 0.3	81.6 $\pm$ 0.8	91.8 $\pm$ 0.1	0.26M
LoRA-4	95.5 $\pm$ 0.2	87.2 $\pm$ 0.7	64.7 $\pm$ 1.2	94.5 $\pm$ 0.1	83.6 $\pm$ 0.4	91.8 $\pm$ 0.1	1.8M
RandLoRA-100	95.5 $\pm$ 0.3	90.1 $\pm$ 0.4	67.4 $\pm$ 0.3	94.1 $\pm$ 0.3	84.5 $\pm$ 0.3	91.4 $\pm$ 0.6	1.8M
<i>Spectral instruments (reported protocol; six seeds)</i>							
UIOrthoLoRA	96.3 $\pm$ 0.3	92.5 $\pm$ 0.2	67.4 $\pm$ 0.5	95.0 $\pm$ 0.1	86.0 $\pm$ 0.2	92.1 $\pm$ 0.2	1.008M †
UILinLoRA	96.3 $\pm$ 0.2	93.4 $\pm$ 0.7	68.7 $\pm$ 1.0	94.7 $\pm$ 0.1	86.0 $\pm$ 1.1	92.0 $\pm$ 0.2	0.221M †

5.6 TASK ADAPTATION AND MASKED-TOKEN BEHAVIOR DIFFER

The fixed language-model probe (Appendix B.12) has original-backbone loss 2.0956 nats. Every arm’s mean probe loss is 0.47–0.84 nats higher, and MIX has the highest mean on both tasks. Its lower task loss therefore accompanies worse masked-token loss on this probe. Measuring both makes the functional outcome concrete: the favorable result is the held-out task prediction loss.

5.7 BROADER TASK RESULTS

Table 3 reports the scaled diagonal and rotated adapters across six GLUE tasks and two RoBERTa sizes. On RoBERTa-base, the rotated form has higher reported scores on four tasks, the diagonal form on one, and they tie on one. On RoBERTa-large, each leads on two tasks and they tie on two. Both constructions support task learning across this evaluation; the advantage of the rotated form varies by task and backbone.

These historical experiments and the imported reference results use their respective training and selection protocols, recorded in Appendix B. They provide context for the adapter family alongside the fresh controlled experiment. Supporting GPT-2 generation results appear in Appendix D.1.

A separate 27-run, five-task archive tests the practical interaction penalty. Cross share falls in all nine paired task/seed settings. The two-sided penalty’s task-mean scores are lower by 0.23–1.03 points, while one-sided control has the highest means on three tasks (Appendix B.6). This earlier experiment separates the consistent reduction in interaction from the task-dependent score response. The fresh experiment adds the size control and held-out loss evaluation.

6 RELATION TO SPECTRAL ADAPTATION METHODS

Choosing directions. LoRA restricts update rank (Hu et al., 2022); AdaLoRA allocates an update’s singular-value budget (Zhang et al., 2023). PiSSA and MiLoRA initialize trainable factors from leading and minor pretrained components (Meng et al., 2024; Wang et al., 2025b). KaSA and CorDA make other decomposition-guided choices (Wang et al., 2025a; Yang et al., 2024). We use a

fixed pretrained reference to distinguish the initial choice from the region occupied by the trained update. Free factors can move beyond their initial spans, while our fixed tail bases enforce confinement throughout training.

Combining directions. SVDiff trains spectral shifts, SVFT learns selected combinations of fixed singular vectors, and LoRA-XS trains a small core on fixed support (Han et al., 2023; Lingam et al., 2024; Bałazy et al., 2025). Spectral Adapter rotates a selected leading core, and PSOFT relaxes orthogonality inside principal support (Zhang & Pilanci, 2024; Wu et al., 2026). Our adapter family brings these choices into one controlled construction. The approximation identity separates flexibility within the chosen region from access to directions outside it; the interaction analysis explains how ambient scaling connects the groups. Appendix A.5 derives the coordinate forms of these methods.

Rotations, scaling and restricted adaptation. OFT/BOFT and HRA use orthogonal transformations (Qiu et al., 2023; Liu et al., 2024b; Yuan et al., 2024). VeRA, RandLoRA, DoRA, SSF and IA3 use other scaling or factor restrictions (Kopiczko et al., 2024; Albert et al., 2025; Liu et al., 2024a; Lian et al., 2022; Liu et al., 2022), while FourierFT selects Fourier coefficients (Gao et al., 2024). Work on restricted parameter spaces motivates studying which freedoms matter for adaptation (Li et al., 2018; Aghajanyan et al., 2020). We connect that question to measured weight changes and task behavior, treating behavioral retention (Shuttleworth et al., 2025) and ranked-subspace perturbation (Wedin, 1972) as separate measurements. Our controlled result compares interaction and size penalties within the same spectral adapter.

7 SCOPE AND PRACTICAL CONSIDERATIONS

The controlled evidence covers one encoder, two tasks, three confirmation seeds and one fixed interaction strength. The loss ordering was identified after inspecting the held-out evaluation. Cross-entropy depends on confidence as well as predicted labels; temperature scaling fitted on separate inner-selection logits is the next check (Appendix B.16). The MIX-NORM accuracy differences have nominal 95% paired intervals $[-3.92, 4.16]$ points on RTE and $[-3.89, 3.24]$ on MRPC. The small observed mean differences therefore describe these runs rather than establish equivalent accuracy.

The comparison measures the full effect of the two training penalties. Confirmation size errors reach 8.0%, with five of six pairs outside the calibration rule’s 5% tolerance, and module allocations differ. The small-update sensitivity analysis conditions on attained norms, a post-treatment variable. Isolating cross share as the cause of the loss difference, or the pretrained projector as the essential ingredient, requires additional controls. The separate 21-run band study uses RTE inner-selection scores; its frozen checkpoints have not yet been evaluated on the official held-out split. It establishes neither tail superiority across tasks nor a convergence advantage. Rotations add parameters as well as flexibility.

The practical base adapter has 86,016 trainable adapter entries and stores 108 MiB of frozen bases at two bytes per entry. Its two dense basis products require about 192 times the branch multiply-accumulates of rank-4 LoRA at width 768; this is an arithmetic comparison. Merging the update removes the adapter branch for inference. Appendix C separates setup, storage, training and merge costs. The historical benchmark protocols and configuration-derived parameter counts remain separately labeled.

8 CONCLUSION

Our adapters separate where a model can adapt from how directions can interact. Strict tail coefficients improve RTE accuracy over a trained head alone; adding rotations within that tail does not give a consistent gain. When both leading and tail updates are allowed, interaction control lowers held-out prediction loss by 44% and 48% relative to the tested size penalty, with mean accuracy differences below half a point. Controlling interaction changes how the model adapts, and accuracy alone misses the resulting difference in prediction loss.

REPRODUCIBILITY STATEMENT

Complete proofs appear in Appendix A, training and metric definitions in Appendix B, and computational costs in Appendix C. Archived records and scripts reproduce the legacy tables; separate exports reproduce the focused dose frontier, confirmation contrasts, module profiles and all 21 band/head outcomes. Synthetic-matrix checks cover the algebra and boundary cases. The supplementary archive, `analysis_artifact.zip`, contains manuscript sources, exported measurements, per-example held-aside predictions, 13 recipe manifests, the recorded experiment implementation, and regeneration/audit scripts with file hashes. Private identifiers are sanitized, with original and distributed hashes distinguished. No model checkpoints or prepared dataset tensors are included; the analysis audit does not rerun training or checkpoint validation. Supporting benchmark tables distinguish imported comparisons from the spectral-variant results.

AI USE STATEMENT

Generative AI tools assisted with conceptual framing, mathematical formulation and proof drafting, literature lookup, experiment design, implementation of experiment and analysis code, interpretation of results, figure and table generation, and manuscript drafting and editing. The reported model measurements come from experimental artifacts; synthetic-matrix checks are identified separately. Automated algebra, data-reconstruction, and document checks support verification but do not replace author review of the scientific claims. The authors are responsible for the final text, code, and reported results.

ETHICS STATEMENT

The study analyzes adaptation of existing language models using existing benchmarks. Weight-space stability does not certify factual correctness, fairness, privacy, or safe behavior. Efficient adaptation can reduce training costs but can also facilitate harmful specialization or preserve biases present in a base model. Our structural claims should not be interpreted as deployment guarantees.

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NOTATION GUIDE

Symbol	Meaning
W_0, W_{eff}	Pretrained and adapted effective weight matrices.
U, Σ, V	Fixed pretrained SVD; L denotes leading and T its tail complement.
d, k, r, q	Square weight width, leading cutoff, tail width $r = d - k$, rotated tail width.
Δ, C	Effective update $W_{\text{eff}} - W_0$ and coordinates $U^\top \Delta V$.
C_{ab}	Block with output subspace a and input subspace b , $a, b \in \{L, T\}$.
Δ_{learned}	Displacement from the inserted model, $W_{\text{eff}} - W_{\text{eff}}(0)$.
S_L, H	Leading and tail cores in the unified additive update; practical $S_L = I$.
E, D	Ambient diagonal output/input scalars.
A, B	In the bounds: $U^\top EU, V^\top DV$; prior-method factor conventions are stated locally.
M_E, M_D	Scaler mixing operators $U_L^\top EU_T, V_L^\top DV_T$.
μ, ν	Operator norms of M_E, M_D .
e, d_s, s, h_s	Operator norms of E, D, S_L, H , respectively.
p_{ab}, p_{cross}	Normalized block energies and $p_{LT} + p_{TL}$.
$p_{\text{off}}, q_{\text{off}}$	Off-tail energy fraction and its square root (norm ratio).
ρ_F	Relative Frobenius update norm; pooling convention specified per table.
η_I	Fraction of leading-block energy in its fitted scalar-identity component.
P, Q	Left/right leading projectors; context specifies any random projector.
d_{ch}, o	Normalized chordal distance and leading-subspace overlap.
λ_E, λ_D	Coefficients of the summed scaler-mixing penalties.
β	Coefficient of the mean module-relative squared-norm penalty.
δ_s, e_s	Signed NORM/MIX norm error and its absolute value for seed s .
$k_{\text{val}}, k_{\text{vec}}$	Implementation settings for coefficient and rotation dimensions.

A PROOFS AND BOUNDARY CASES

A.1 FIXED COEFFICIENTS AND WITHIN-TAIL FLEXIBILITY

Let $\hat{G} = U^\top GV$ and partition it into the four coordinate blocks. Orthogonal invariance gives

$$\|G - U_T H V_T^\top\|_F^2 = \|\hat{G}_{LL}\|_F^2 + \|\hat{G}_{LT}\|_F^2 + \|\hat{G}_{TL}\|_F^2 + \|\hat{G}_{TT} - H\|_F^2. \quad (14)$$

For a free core the last term is minimized at $H = \hat{G}_{TT}$. For a diagonal core it is minimized by $h_i = (\hat{G}_{TT})_{ii}$, leaving the off-diagonal entries unchanged. Substitution proves Proposition 1. An SVD of the minimizing free core gives the representation (3) with $q = r$. Thus full within-tail rotations can attain the free-core approximation optimum.

For a separate family whose free core has rank at most $q \leq r$, rather than the partial-rotation core of (3), truncated SVD of $\hat{G}_{TT}$ gives minimum squared error

$$\|G\|_F^2 - \sum_{j=1}^q \sigma_j(\hat{G}_{TT})^2. \quad (15)$$

This expression keeps the outside-tail residual explicit. It is the optimum under the fixed-tail restriction; the unrestricted rank- q approximation of G can use directions elsewhere.

A.2 PROOF OF THE UNIFIED MIXING BOUNDS

Proposition 2 (Interaction bounds with and without a leading core). *Write $e = \|E\|_2$, $d_s = \|D\|_2$, $s = \|S_L\|_2$, and $h_s = \|H\|_2$. For (7),*

$$\|C_{LL}\|_2 \leq esd_s + \mu h_s \nu, \quad \|C_{LT}\|_2 \leq es\nu + \mu h_s d_s, \quad (16)$$

$$\|C_{TL}\|_2 \leq \mu s d_s + e h_s \nu, \quad \|C_{TT}\|_2 \leq \mu s \nu + e h_s d_s. \quad (17)$$

If $\mu = \nu = 0$, the cross blocks vanish. In addition, the leading block vanishes when $S_L = 0$.

The proof applies submultiplicativity to (10). The additional leading-core terms are the difference between the idealized and practical analyses. With $S_L = 0$, leading–leading leakage requires mixing on both sides, while each cross block can be first order in one mixing quantity. With $S_L \neq 0$, direct leading adaptation remains even when the cross terms vanish.

Corollary 3 (Tail confinement and block decoupling). *For $S_L = 0$, define $C_{\text{tail}} = \text{diag}(0, C_{TT})$. Then*

$$\|C - C_{\text{tail}}\|_2 \leq h_s \sqrt{(\mu\nu)^2 + (\mu d_s)^2 + (e\nu)^2}. \quad (18)$$

Consequently, $\mu, \nu \rightarrow 0$ implies confinement in absolute block norm if e, d_s, h_s stay bounded. If e, d_s, h_s, s stay bounded for a general leading core, the same limit instead makes C block-diagonal; it need not make C_{LL} small.

Because U, V are orthogonal,

$$C = A \begin{bmatrix} S_L & 0 \\ 0 & H \end{bmatrix} B. \quad (19)$$

Direct multiplication gives

$$C_{LL} = A_{LL} S_L B_{LL} + A_{LT} H B_{TL}, \quad (20)$$

$$C_{LT} = A_{LL} S_L B_{LT} + A_{LT} H B_{TT}, \quad (21)$$

$$C_{TL} = A_{TL} S_L B_{LL} + A_{TT} H B_{TL}, \quad (22)$$

$$C_{TT} = A_{TL} S_L B_{LT} + A_{TT} H B_{TT}. \quad (23)$$

Symmetry yields $\|A_{TL}\|_2 = \|A_{LT}\|_2 = \mu$ and $\|B_{TL}\|_2 = \|B_{LT}\|_2 = \nu$. Orthogonal compression bounds the diagonal blocks of A by e and those of B by d_s . Apply the triangle inequality and submultiplicativity to each product to obtain (16)–(17).

For the tail-only core, $s = 0$ and

$$C - C_{\text{tail}} = \begin{bmatrix} C_{LL} & C_{LT} \\ C_{TL} & 0 \end{bmatrix}. \quad (24)$$

For any block vector (x, y) , the norms of the two output blocks are bounded componentwise by

$$\begin{bmatrix} \|C_{LL}\|_2 & \|C_{LT}\|_2 \\ \|C_{TL}\|_2 & 0 \end{bmatrix} \begin{bmatrix} \|x\|_2 \\ \|y\|_2 \end{bmatrix}. \quad (25)$$

Taking a supremum and bounding the norm of this nonnegative 2×2 matrix by its Frobenius norm proves

$$\|C - C_{\text{tail}}\|_2 \leq \sqrt{\|C_{LL}\|_2^2 + \|C_{LT}\|_2^2 + \|C_{TL}\|_2^2}. \quad (26)$$

Substitution gives (18). If the stated factor norms remain bounded, its right-hand side goes to zero as $\mu, \nu \rightarrow 0$.

For the general core, let $C_{\text{diag}} = \text{diag}(C_{LL}, C_{TT})$. The off-diagonal block matrix satisfies the exact identity

$$\|C - C_{\text{diag}}\|_2 = \max\{\|C_{LT}\|_2, \|C_{TL}\|_2\}. \quad (27)$$

The cross-block bounds therefore imply convergence to the set of block-diagonal updates when e, d_s, h_s, s are bounded and $\mu, \nu \rightarrow 0$. The term $A_{LL}S_LB_{LL}$ need not vanish. This proves Corollary 3 and states precisely the different limits in the two cases.

A.3 CONFINEMENT DOES NOT PREVENT SINGULAR-VALUE CROSSINGS

For a tail-only update without ambient scaling,

$$U^\top(W_0 + \Delta)V = \begin{bmatrix} \Sigma_L & 0 \\ 0 & \Sigma_T + H \end{bmatrix}. \quad (28)$$

The singular values are the union of those of the two diagonal blocks, which proves the sufficient condition (6). Consider $W_0 = \text{diag}(2, 1)$, cutoff $k = 1$, and $\Delta = \text{diag}(0, 2)$. The update is strictly tail-supported with no cross mixing and no leading-block modification, but $W_0 + \Delta = \text{diag}(2, 3)$ has the other coordinate as its top singular direction. Its largest-principal-angle sine relative to the original leading direction is one. Thus block decoupling and stability of the top-ranked subspace are distinct even in a two-dimensional example.

A.4 RECTANGULAR WEIGHTS AND COMPLETE ENERGY ACCOUNTING

For $W_0 \in \mathbb{R}^{m \times n}$, select $k \leq \min(m, n)$ leading singular vectors and define

$$P_L = U_L U_L^\top, \quad P_T = I_m - P_L, \quad Q_L = V_L V_L^\top, \quad Q_T = I_n - Q_L. \quad (29)$$

Then $\Delta_{ab} = P_a \Delta Q_b$ gives

$$\Delta = \sum_{a,b} \Delta_{ab}, \quad \|\Delta\|_F^2 = \sum_{a,b} \|\Delta_{ab}\|_F^2. \quad (30)$$

Pairwise Frobenius inner products vanish because complementary projectors multiply to zero. The tail here is the full complement and includes any unmatched null directions. A chosen adapter may span only a subset of it; that available subspace must not be confused with the diagnostic complement. Complete orthogonal bases recover the corresponding coordinate blocks. The square attention-matrix experiment avoids this issue, but an economy-SVD diagnostic is not automatically exhaustive for other layers.

A.5 REPRESENTATIVE METHODS IN THE FIXED COORDINATE FRAME

These are algebraic consequences of update forms, not measurements on prior methods. Let Q_J select the declared pretrained singular indices J , and write $A_a = U_a^\top A$, $B_b = B V_b$. Any additive update AB has $C_{ab} = A_a B_b$; low rank alone forces no LL/LT/TL/TT block to vanish. A learned sum of low-rank terms has the sum of these blocks. A fixed-basis core instead has $\Delta = U Q_J K Q_J^\top V^\top$ and $C = Q_J K Q_J^\top$, so only its selected support is reachable.

For LoRA and AdaLoRA the low-rank factors are trainable (Hu et al., 2022; Zhang et al., 2023). PiSSA/MiLoRA’s retained residual means displacement from the original checkpoint subtracts the initial product; the leading/minor initialization does not freeze the factor spans (Meng et al., 2024;

Table 4: Representative coordinate placements, relative to the original W_0 . “All” means no block is structurally forced to zero, not that every fitted update populates every block. The formulas absorb any fixed adapter scaling into the coefficients.

Method/form	Effective update	Coordinates C	Permitted support
LoRA / AdaLoRA	AB / APB	$U^\top ABV / U^\top APBV$	All
PiSSA / MiLoRA	$AB - A_0B_0$	$U^\top (AB - A_0B_0)V$	All after free-factor training
SVDiff	$U \text{diag}(h)V^\top$	$\text{diag}(h)$	LL, TT diagonal
SVFT	$UMV^\top$	M	Declared sparsity pattern
LoRA-XS	$U_J \Sigma_J R V_J^\top$	$Q_J \Sigma_J R Q_J^\top$	$J \times J$
Spectral Adapter (rotation)	$U_L(R_U \Sigma_L R_V^\top - \Sigma_L)V_L^\top$	leading core difference	LL
PSOFT	$A'(K_P - I)B'$	$Q_J(K_P - I)\Sigma_J Q_J^\top$	$J \times J$ (principal)
OFT / BOFT (pure left map)	$(R - I)W_0$	$(U^\top R U - I)\Sigma$	All
HRA (pure right map)	$W_0(R - I)$	$\Sigma(V^\top R V - I)$	All

Wang et al., 2025b). SVDiff changes singular values (Han et al., 2023); SVFT explicitly permits off-diagonal entries (Lingam et al., 2024). In LoRA-XS the published initialization includes Σ_J in the left fixed factor, hence $C_{JJ} = \Sigma_J R$, not literally R unless it is reparameterized (Bałazy et al., 2025). If J lies inside the declared leading set, every other diagnostic block is zero.

The Spectral Adapter entry describes its rotational leading-core variant (Zhang & Pilanci, 2024). OFT/BOFT and HRA use orthogonal R in the pure forms shown (Qiu et al., 2023; Liu et al., 2024b; Yuan et al., 2024); singular-value preservation is compatible with nonzero cross blocks. Left/right placement follows the stated weight convention. Additional learned magnitude scales, such as rescaled OFT, invalidate the pure orthogonal spectrum-preservation statement. These examples distinguish fixed support, a coefficient pattern, and an ambient isometry without equating any of them with preserved behavior. They do not establish cross-method empirical effects.

For PSOFT, $A' = U_J$, $B' = \Sigma_J V_J^\top$, and $K_P = \text{diag}(\alpha)R \text{diag}(\beta)$ act inside a fixed principal core; the residual of W_0 stays frozen (Wu et al., 2026). Thus $C_{JJ} = (K_P - I)\Sigma_J$, with zero support outside $J \times J$ even after vector relaxation. Its geometry condition $R^\top A^\top A R = A^\top A$ preserves column Gram geometry; choosing $A' = U_J$ reduces it to orthogonality. Our Corollary 3 instead bounds support outside specified blocks. Neither property alone implies the other: a nonorthogonal core remains confined, while an ambient orthogonal map can mix blocks. Principal versus tail selection is a further independent choice.

GaLore projects gradients into low-rank subspaces that can change during training (Zhao et al., 2024); the sum of such updates need not remain inside one fixed pretrained span. GOFT/qGOFT use Givens rotations and relaxed angular/norm constraints (Ma et al., 2024), complementing the ambient-transform examples above. These differ from a fixed update-support constraint.

A.6 A DIMENSION-ONLY REFERENCE FOR ALLOCATION

Let the vectorized nonzero update have a rotationally invariant direction in the d^2 -dimensional matrix space. By symmetry each squared coordinate has expected fraction $1/d^2$ of the total. The leading, cross, and tail blocks have k^2 , $2kr$, and r^2 coordinates, respectively. Their expected energy fractions are therefore k^2/d^2 , $2kr/d^2$, and r^2/d^2 . For $r/d = 1/3$, these are $4/9$, $4/9$, and $1/9$, with off-tail expectation $8/9$. The square root $\sqrt{8/9} \simeq 0.9428$ is a reference on the norm-ratio scale, not the expectation $\mathbb{E}[q_{\text{off}}]$. These calculations are a null reference, not a fitted model or a statistical test of the observed updates.

A.7 THE SCALER PENALTY AS A PROJECTOR COMMUTATOR

In the complete U basis, P is $\text{diag}(I_k, 0)$. Writing $A = U^\top E U$ gives

$$U^\top (EP - PE)U = \begin{bmatrix} 0 & -A_{LT} \\ A_{TL} & 0 \end{bmatrix}. \quad (31)$$

Because A is symmetric, orthogonal invariance yields $\|EP - PE\|_F^2 = 2\|A_{LT}\|_F^2$. In ambient coordinates, the (i, j) entry of the commutator is $(\epsilon_i - \epsilon_j)P_{ij}$. Squaring and combining its symmetric

pairs proves (12). Equivalently, the penalty is $\epsilon^\top L_P \epsilon$, where L_P is the graph Laplacian with off-diagonal edge weights P_{ij}^2 . Every summand is nonnegative. Zero penalty therefore means equal scales across every nonzero edge, or constant scales on each connected component. A connected graph has only globally constant scales in this nullspace. A disconnected projector permits different constants on disconnected components; for a coordinate-aligned projector, all off-diagonal weights vanish and every ambient diagonal scaler commutes with it. The result is conditional on the fixed pretrained projector, not a measured connectivity claim for the experiment.

A.8 RANDOM-PROJECTOR REFERENCE AND SCALER HOMOGENIZATION

Write $J_P(\epsilon) = \epsilon^\top L_P \epsilon$ for (12). Since $L_P \mathbf{1} = 0$, adding a constant to every scale does not change the penalty. Under the penalty alone, $\dot{\epsilon} = -2\lambda L_P \epsilon$ preserves $\bar{\epsilon}$ and damps each positive-eigenvalue component of L_P . On a connected graph it converges to $\bar{\epsilon} \mathbf{1}$. This statement concerns ordinary gradient flow of this isolated quadratic, not AdamW with a task loss; neither shrinkage of the mean nor convergence of the trained model follows from it.

For a uniformly random rank- k orthogonal projector P in $\mathbb{R}^d$, and a fixed scaler vector independent of P ,

$$\mathbb{E}_P[J_P(\epsilon)] = \frac{k(d-k)}{(d-1)(d+2)} \|\epsilon - \bar{\epsilon} \mathbf{1}\|_2^2. \quad (32)$$

To verify the constant, write $P = QQ^\top$ for the first k columns of a Haar orthogonal matrix. The spherical fourth moments give $\mathbb{E}[P_{ii}^2] = k(k+2)/(d(d+2))$. Since $\sum_{ij} P_{ij}^2 = k$, symmetry gives $\mathbb{E}[P_{ij}^2] = k(d-k)/(d(d-1)(d+2))$ for $i \neq j$. Substitute into (12) and use $\sum_{i < j} (\epsilon_i - \epsilon_j)^2 = d\|\epsilon - \bar{\epsilon} \mathbf{1}\|_2^2$.

This expectation motivates a centered-scaler control. It does not equate one fixed random projector with the pretrained projector: their positive-eigenvalue directions and strengths generally differ. It also cannot be substituted into training when the learned scalars depend on the sampled projector. A shared scalar nullspace alone does not imply identical trajectories or endpoints.

A.9 FACTOR-SCALE IDENTITIES

Write the inner matrix in (7) as K . For any $a > 0$, $(aE)K(D/a) = EKD$. Because each mixing operator is linear in its respective scaler, this transformation multiplies its two magnitudes by a and $1/a$. For $S_L = 0$, the additional identity

$$(aE)U_T(H/a^2)V_T^\top(aD) = EU_T H V_T^\top D \quad (33)$$

shrinks both squared Frobenius penalties by a^2 . It can make a raw factor penalty arbitrarily small with a fixed effective update if the core is allowed to diverge. This rescaling illustrates why the confinement bound assumes bounded factors. Measuring the effective update and the factor norms checks both sides of that statement.

A.10 MAGNITUDE FLOW IN EFFECTIVE AND FACTOR COORDINATES

For a freely optimized effective update $x = \text{vec}(\Delta)$, isolated gradient flow of $\beta\|x\|_2^2$ is radial, $\dot{x} = -2\beta x$, and preserves block fractions. If $x = f(\theta)$ is parameterized by adapter factors with Jacobian J_θ , ordinary parameter gradient flow instead induces

$$\dot{x} = -2\beta J_\theta J_\theta^\top x. \quad (34)$$

This need not be radial. A task objective or adaptive optimizer introduces further changes. For an explicit penalty-only example, fix $E = \text{diag}(2, 1)$, $D = V = I$, let U rotate by $\pi/6$, fix $S_L = 1$ and train only $H = h$. Then

$$C = \begin{bmatrix} 7/4 & -\sqrt{3}h/4 \\ -\sqrt{3}/4 & 5h/4 \end{bmatrix}, \quad \|C\|_F^2 = (13 + 7h^2)/4, \quad p_{\text{cross}} = \frac{3(1+h^2)}{4(13+7h^2)}. \quad (35)$$

Magnitude-only flow decreases h and changes p_{cross} (0.075 at $h = 1$, approaching $3/52$ at zero). Thus orthogonal invariance of the norm objective does not force a constant trained cross share. On the MIX side, connected-projector, penalty-only homogenization predicts a scalar limit; finite task-trained cross shares and identity residuals remain empirical quantities.

B EXPERIMENTAL DETAILS

B.1 BACKBONES, TASKS, AND PLACEMENT

The broader language-understanding evaluation uses RoBERTa-base and RoBERTa-large on a six-task GLUE subset: CoLA, SST-2, MRPC, STS-B, QNLI, and RTE (Wang et al., 2018). The archived mixing intervention uses RoBERTa-base on five of these tasks, excluding QNLI. Neither covers MNLI or QQP. Table 3 reports Matthews correlation for CoLA, accuracy for SST-2/MRPC/QNLI/RTE, and Pearson correlation for STS-B. The archived mixing intervention instead reports MRPC F1. These historical evaluations use official validation splits; the new fixed-endpoint study uses an inner split and evaluates official validation only after freezing its confirmation checkpoints (Appendix B.8). Backbone parameters are frozen; adapters and the task head train jointly. The adapted matrices are the query, key, value, and attention-output projections; feed-forward projections remain frozen.

B.2 HISTORICAL TRAINING CONFIGURATION

The historical recipe uses AdamW with separate adapter and head learning rates, linear decay, a 6% warmup fraction, batch size 64, and maximum sequence length 256. Table 5 gives the recorded RoBERTa-base settings. The fresh controlled experiment uses its own common recipe, given in Appendix B.12. Imported benchmark methods retain their source protocols.

Table 5: Historical RoBERTa-base training settings. The scaler value initializes both ambient diagonal scalers. The new fixed-endpoint study has a separate protocol (Appendix B.8).

Task	Epochs	Head LR	Adapter LR	Scaler	Coefficients
CoLA	80	$5 \cdot 10^{-3}$	$3 \cdot 10^{-2}$	10^{-2}	10^{-1}
SST-2	40	10^{-2}	$4 \cdot 10^{-2}$	10^{-1}	10^{-1}
MRPC	30	10^{-3}	$5 \cdot 10^{-2}$	10^{-1}	10^{-1}
STS-B	60	$5 \cdot 10^{-3}$	10^{-2}	10^{-1}	10^{-1}
QNLI	25	10^{-3}	$2 \cdot 10^{-2}$	10^{-1}	10^{-1}
RTE	90	$5 \cdot 10^{-4}$	10^{-2}	10^{-2}	10^{-2}

Adapter dropout is zero and the nominal adapter scaling coefficient is one. The broad GLUE configurations use $k_{\text{val}} = 256$, with $k_{\text{vec}} = 0$ for UIInLoRA and 64 for UIOrthoLoRA. The mixing study uses 256 and zero, respectively. Compact rotations, when enabled, use an orthogonal parameterization; only the selected sub-block is rotated. With nonzero initialized scalers and coefficients, the practical adapter need not have zero initial delta. The measured displacement from W_0 includes that initial contribution.

B.3 INITIALIZATION AND RESIDUAL LEADING STRUCTURE

The fixed choice $S_L = I_k$ gives equal inner coefficients to the leading directions, independent of their pretrained singular values. Choosing $S_L = \Sigma_L$ would instead weight this contribution by the pretrained spectrum. All practical conditions use the identity choice; the experiment does not compare these two designs.

For the prescribed identity rotations, $E = D = s_0 I$, $H = h_0 I_r$, and $S_L = I_k$, insertion adds

$$\Delta_{\text{init}} = s_0^2 (U_L V_L^\top + h_0 U_T V_T^\top), \quad \|\Delta_{\text{init}}\|_F = s_0^2 \sqrt{k + r h_0^2}, \quad p_{LL, \text{init}} = \frac{k}{k + r h_0^2}. \quad (36)$$

The cross blocks are initially zero. With $k = 512$, $r = 256$, the prescribed norm is approximately 0.22684 for MRPC/STS-B/SST-2, 0.0022684 for CoLA, and 0.0022628 for RTE. Leading energy is approximately 99.50% in the first four tasks and 99.995% in RTE. These are analytical values from Table 5, not recovered checkpoint measurements. These recipe calculations alone do not quantify the offset relative to each archived trained update. The fresh confirmation export separately supplies measured initial and learned deltas.

For an attained leading block, define its identity alignment

$$\eta_I = \frac{(\text{tr } C_{LL})^2}{k \|C_{LL}\|_F^2}, \quad 0 \leq \eta_I \leq 1 \quad (C_{LL} \neq 0). \quad (37)$$

It equals the fraction of leading-block Frobenius energy in the projection onto $\text{span}(I_k)$. Under constant scalers and $S_L = I_k$, $C_{LL} = abI_k$ has $\eta_I = 1$ when $ab \neq 0$, even though cross mixing vanishes. This motivates reporting $C - C_{\text{init}}$, $C_{LL} - (\text{tr } C_{LL}/k)I_k$, and centered scaler norms alongside total block energies. Subtracting the initialization and subtracting a fitted identity component answer different questions; neither can be reconstructed by subtracting squared norms. The fresh confirmation export supplies identity alignment and residual energies, analyzed in Appendix B.13; centered scaler vectors remain unavailable. The older archive lacks these trained measurements.

B.4 SEEDS AND CHECKPOINT SELECTION

The broader spectral evaluation protocol reports six seeds $\{42, 123, 2021, 17, 31415, 1054\}$; Table 3 retains its reported means and across-seed standard deviations. The imported comparison uses five runs. The mixing analysis uses the nine task/seed pairs with archived paired summary and module-level records: seeds 42 and 17 for RTE, MRPC, CoLA, and STS-B, and 42 for SST-2. These records permit independent reaggregation of the intervention, whereas the broad GLUE table is a report of the original evaluation, not a newly reconstructed six-seed experiment.

The archived ablation runner selects the best validation checkpoint by the reported task metric, then reloads it for scoring and spectral measurements. Table 7 records each selected step; RTE seed-42 A/B/C select steps 312, 3198, and 1755. These are selection-set outcomes. Table 6 gives sample SD for the two-seed settings and individual values for the one-seed setting. The fresh study instead fixes training endpoints before held-out evaluation.

B.5 METRIC DEFINITIONS AND AGGREGATION

For each adapted matrix, the reported metrics are

$$\rho_F = \|\Delta\|_F / \|W_0\|_F, \quad q_{\text{off}} = \|C_{\text{off}}\|_F / \|C\|_F, \quad (38)$$

$$\text{Leak}_{LL} = \|C_{LL}\|_2 / \|\Delta\|_2, \quad C_{\text{off}} = \begin{bmatrix} C_{LL} & C_{LT} \\ C_{TL} & 0 \end{bmatrix}. \quad (39)$$

The mixing magnitudes are operator norms, while their penalties use squared Frobenius norms. Leading left and right drift use the largest-principal-angle sine between the leading subspaces of W_0 and $W_0 + \Delta$. The archived tables average each metric across the 48 modules within a run, then across seeds; they do not pool module energies before normalization.

The new fixed-endpoint study instead reports

$$\rho_{F,\text{pool}} = \sqrt{\frac{\sum_{\ell} \|\Delta_{\ell}\|_F^2}{\sum_{\ell} \|W_{0,\ell}\|_F^2}}, \quad p_{ab,\text{pool}} = \frac{\sum_{\ell} \|C_{ab,\ell}\|_F^2}{\sum_{\ell} \|\Delta_{\ell}\|_F^2}. \quad (40)$$

Its equal-module comparison averages the individual $p_{ab,\ell}$ instead. A pooled fraction weights modules by update energy; an equal-module fraction does not. NORM’s training objective averages squared module-relative norms, whereas calibration targets $\rho_{F,\text{pool}}$. Neither matching a pooled norm nor rescaling it matches each module. Zero-update fractions are undefined. In both protocols, seeds, not modules, are the replication units.

B.6 ARCHIVED INTERVENTION: DESIGN AND SUMMARY

The archived intervention uses RoBERTa-base (Liu et al., 2019) on RTE, MRPC, CoLA, STS-B, and SST-2. Adapters modify the query, key, value, and attention-output projections in each of its 12 layers, giving 48 adapted matrices. The configuration adapts 256 coefficients with $k_{\text{vec}} = 0$, holding rotations out of the comparison. The analysis uses the practical leading-plus-tail form (7). The three conditions are A: no penalty, B: $\lambda_E = 10^{-3}$, $\lambda_D = 0$, and C: $\lambda_E = \lambda_D = 10^{-3}$. The backbone is frozen and the task head trains jointly with the adapter.

Table 6: All three conditions: A unregularized, B left-only, C two-sided. Entries are mean $\pm$ sample SD across n seeds; $n = 1$ is a single run, with no estimated dispersion. Scores and cross fractions are multiplied by 100. Spectral metrics first average 48 modules within each selected checkpoint. Scores use MRPC F1 here, unlike Table 3. Per-seed values and selection steps appear in Appendix B.7.

Task	Condition	n	Task score	ρ_F	Cross energy (%)
RTE	A	2	77.08 $\pm$ 1.28	0.3787 $\pm$ 0.0142	43.41 $\pm$ 0.15
RTE	B	2	79.06 $\pm$ 1.02	0.2847 $\pm$ 0.0212	43.55 $\pm$ 0.43
RTE	C	2	76.71 $\pm$ 0.77	0.0626 $\pm$ 0.0004	27.45 $\pm$ 1.93
MRPC	A	2	92.15 $\pm$ 0.45	0.3174 $\pm$ 0.0571	43.25 $\pm$ 0.14
MRPC	B	2	92.40 $\pm$ 0.06	0.1958 $\pm$ 0.0552	40.32 $\pm$ 2.13
MRPC	C	2	91.32 $\pm$ 0.04	0.0430 $\pm$ 0.0109	10.08 $\pm$ 3.47
CoLA	A	2	64.09 $\pm$ 1.40	0.3076 $\pm$ 0.0048	43.75 $\pm$ 0.10
CoLA	B	2	64.30 $\pm$ 1.11	0.2057 $\pm$ 0.0041	42.38 $\pm$ 0.23
CoLA	C	2	63.14 $\pm$ 1.22	0.0334 $\pm$ 0.0208	21.09 $\pm$ 5.63
STS-B	A	2	90.80 $\pm$ 0.27	0.3820 $\pm$ 0.0030	43.46 $\pm$ 0.01
STS-B	B	2	90.48 $\pm$ 0.13	0.2506 $\pm$ 0.0702	41.96 $\pm$ 0.23
STS-B	C	2	90.58 $\pm$ 0.39	0.0679 $\pm$ 0.0128	13.71 $\pm$ 1.36
SST-2	A	1	94.72	0.2820	43.33
SST-2	B	1	94.38	0.2676	40.19
SST-2	C	1	93.69	0.0557	17.21

The analysis comprises seeds 42 and 17 for RTE, MRPC, CoLA, and STS-B, and seed 42 for SST-2: nine paired task/seed comparisons. Spectral quantities are measured at each run’s validation-selected checkpoint, averaged over its 48 matrices, then averaged over seeds within a task. Runs, rather than layers, are the units of replication. Checkpoint selection is shared as a rule, not necessarily as an optimizer step. Appendix B gives the training settings, metric definitions, and complete three-condition results.

The diagnostic cutoff is $k = 512$ in each 768×768 matrix, leaving the 256 adapted tail directions. Table 6 reports task score, relative norm ρ_F , and cross energy. Appendix B.7 retains q_{off} and both largest-principal-angle sines. The latter saturate near one under A: a single boundary direction can cause this, so they are supplementary indicators, not measures of how much the entire leading subspace moved.

B.7 COMPLETE MIXING RESULTS

The accompanying `scripts/build_mixing_tables.py` generates Tables 6 and 7 from the included per-task/seed CSV summaries. These records retain checkpoint steps and metric columns. The analysis includes the paired A/B/C runs and excludes separate smoke tests and C-only reruns. This makes the aggregation independently checkable without rerunning training.

B.8 MAGNITUDE CONTROL: CALIBRATION AND CONFIRMATION

The study of Section 5.2 follows registered protocols with file hashes. Each included run passed checkpoint reload and reproduction of task metrics, spectral diagnostics and the frozen-head probe. The common recipe uses sequence length 128, effective batch 32, an inner training/selection split, and fixed endpoints of 5,670 steps for RTE and 2,760 for MRPC. Calibration uses seed 31415; confirmation uses seeds 17, 42 and 123.

The registered protocol fixes $\lambda_E = \lambda_D = 10^{-3}$ on both tasks, the same coefficients as the archived two-sided condition. The matching search varies only NORM’s coefficient; it does not select MIX’s dose from confirmation outcomes.

The dose-selection rule targets MIX’s pooled relative Frobenius norm of the total effective update, including insertion. It selects the smallest absolute relative error, with ties toward the lower dose,

Table 7: Every archived mixing run and its validation-selected step. A: no penalty; B: left-only; C: two-sided. Scores are in native metric units (MRPC F1). Spectral values are means over 48 modules of that run: μ_E, ν_D and drift use operator norms, while ρ_F, q_{off} are Frobenius norm ratios. Drift is the largest-angle sine, often near its ceiling.

Task	Cond.	Seed	Step	Score	μ_E	ν_D	ρ_F	q_{off}	Drift _U	Drift _V
RTE	A	17	6318	0.7798	1.392	1.464	0.389	0.948	1.000	1.000
RTE	A	42	312	0.7617	1.376	1.420	0.369	0.947	1.000	1.000
RTE	B	17	3510	0.7978	0.331	2.173	0.300	0.934	0.999	0.999
RTE	B	42	3198	0.7834	0.225	2.051	0.270	0.933	1.000	1.000
RTE	C	17	3588	0.7726	0.208	0.201	0.062	0.823	0.534	0.523
RTE	C	42	1755	0.7617	0.234	0.207	0.063	0.853	0.511	0.500
MRPC	A	17	1725	0.9247	1.347	1.405	0.358	0.947	1.000	1.000
MRPC	A	42	1102	0.9183	1.217	1.284	0.277	0.948	0.998	0.999
MRPC	B	17	2300	0.9236	0.242	1.714	0.235	0.941	1.000	1.000
MRPC	B	42	696	0.9244	0.276	1.361	0.157	0.955	0.966	0.966
MRPC	C	17	1495	0.9129	0.226	0.182	0.035	0.865	0.606	0.601
MRPC	C	42	1334	0.9134	0.234	0.193	0.051	0.848	0.728	0.723
CoLA	A	17	13400	0.6508	1.248	1.325	0.304	0.946	0.999	0.999
CoLA	A	42	4556	0.6310	1.250	1.346	0.311	0.946	0.999	0.999
CoLA	B	17	4556	0.6351	0.288	1.631	0.209	0.937	0.983	0.983
CoLA	B	42	4556	0.6508	0.285	1.605	0.203	0.939	0.995	0.995
CoLA	C	17	14472	0.6401	0.153	0.165	0.048	0.860	0.649	0.643
CoLA	C	42	6432	0.6228	0.166	0.123	0.019	0.900	0.468	0.460
STS-B	A	17	8820	0.9099	1.355	1.450	0.384	0.947	1.000	1.000
STS-B	A	42	9720	0.9061	1.395	1.443	0.380	0.947	1.000	0.999
STS-B	B	17	9360	0.9039	0.231	1.778	0.300	0.936	1.000	1.000
STS-B	B	42	2340	0.9057	0.319	1.665	0.201	0.943	1.000	1.000
STS-B	C	17	8820	0.9030	0.200	0.256	0.059	0.816	0.712	0.709
STS-B	C	42	9540	0.9085	0.203	0.240	0.077	0.795	0.786	0.784
SST-2	A	42	71570	0.9472	1.219	1.255	0.282	0.948	1.000	1.000
SST-2	B	42	50520	0.9438	0.185	1.604	0.268	0.943	1.000	1.000
SST-2	C	42	46310	0.9369	0.134	0.170	0.056	0.850	0.679	0.672

Table 8: Confirmation of the calibration-selected norm control across seeds 17, 42, and 123 (mean±sample SD, $n = 3$), fixed-step endpoints, common protocol of Section 5.2. Cross P, LL, and TT are pooled energy percentages; Cross E weights modules equally. The probe is masked-token cross-entropy under the frozen original MLM head; accuracy is the inner-selection split at the fixed step, scaled by 100. Achieved errors $\delta_s = \rho_{\text{NORM},s}/\rho_{\text{MIX},s} - 1$ (signed; $e_s = |\delta_s|$) against the declared $\pm 5\%$ rule, in that seed order: RTE: +8.0, −5.2, +7.5%; MRPC: +5.5, +6.7, −0.1%. N is the dimension-only reference, not a run or test of isotropy.

Task	Condition	ρ_F	Cross P	Cross E	LL	TT	Probe CE	Acc
RTE	UNREG	0.2751 ± 0.0041	43.7 ± 0.0	43.5 ± 0.0	45.9 ± 0.1	10.4 ± 0.0	2.89 ± 0.17	72.4 ± 0.6
RTE	MIX	0.0736 ± 0.0046	10.0 ± 0.5	14.9 ± 0.9	45.2 ± 0.9	44.8 ± 1.4	2.94 ± 0.29	72.6 ± 3.1
RTE	NORM ($\beta = 7.5$)	0.0759 ± 0.0016	46.2 ± 0.0	46.3 ± 0.1	40.7 ± 0.0	13.1 ± 0.0	2.61 ± 0.08	74.6 ± 0.5
MRPC	UNREG	0.1802 ± 0.0009	42.4 ± 0.1	42.2 ± 0.0	48.0 ± 0.1	9.6 ± 0.1	2.75 ± 0.07	84.9 ± 0.2
MRPC	MIX	0.0407 ± 0.0017	4.5 ± 0.3	5.5 ± 0.1	69.8 ± 1.1	25.8 ± 1.4	2.84 ± 0.17	84.5 ± 0.5
MRPC	NORM ($\beta = 27$)	0.0423 ± 0.0005	46.2 ± 0.1	46.2 ± 0.1	40.6 ± 0.2	13.2 ± 0.1	2.57 ± 0.11	84.8 ± 0.2
—	N (dimension only)	—	44.4	44.4	44.4	11.1	—	—

using validated fixed-step norms. The target tolerance is $\pm 5\%$. The initial grid is $\beta \in \{0.01, 1, 100\}$ per task. A separately registered norms-only refinement adds bracketing geometric means and log-linear-fit-guided doses. The export retains every tried dose, immutable selection records and per-module norms. For each confirmation seed, the reported error is $e_s = |\rho_{\text{NORM},s}/\rho_{\text{MIX},s} - 1|$.

Table 9: Complete Frobenius-control dose frontier at the fixed-step endpoint, calibration seed 31415. Error is the absolute relative deviation of the pooled norm from the mixing arm’s target (RTE 0.0792, MRPC 0.0413); Cross and TT are pooled energy percentages; the probe is masked-token cross-entropy. Every tried dose is shown; none is omitted.

Task	β	ρ_F	Error (%)	Cross (%)	TT (%)	Probe CE
RTE	0.01	0.2766	249.1	43.7	10.4	2.98
RTE	1	0.1621	104.6	44.4	11.1	2.62
RTE	4	0.0976	23.2	45.5	12.2	2.58
RTE	5.5	0.0856	8.0	45.7	12.6	2.54
RTE	7.5	0.0760	4.1	46.1	13.0	2.55
RTE	10	0.0654	17.4	46.5	13.4	2.67
RTE	100	0.0229	71.2	50.9	19.0	2.79
MRPC	0.01	0.1793	334.1	42.4	9.8	2.59
MRPC	1	0.1380	234.2	42.8	10.1	2.53
MRPC	10	0.0667	61.6	44.8	11.8	2.55
MRPC	14	0.0575	39.2	45.3	12.4	2.60
MRPC	19.5	0.0498	20.6	45.7	12.9	2.57
MRPC	27	0.0422	2.3	46.3	13.5	2.71
MRPC	100	0.0212	48.7	48.6	17.1	2.77

Table 10: All confirmation task outcomes on the inner-selection split, scaled by 100. Fixed steps are 5,670 (RTE) and 2,760 (MRPC). Selected checkpoints are secondary, selection-conditioned outcomes on the same split; the main comparisons use fixed endpoints. F1 is shown for MRPC only.

Task	Arm	Seed	Fixed acc.	Fixed F1	Selected step	Selected acc.	Selected F1
RTE	UNREG	17	71.69	—	1920	73.49	—
RTE	UNREG	42	72.89	—	1280	75.70	—
RTE	UNREG	123	72.49	—	1152	76.10	—
RTE	MIX	17	75.90	—	4224	76.10	—
RTE	MIX	42	71.89	—	4253	73.49	—
RTE	MIX	123	69.88	—	2816	73.69	—
RTE	NORM	17	74.90	—	1418	75.50	—
RTE	NORM	42	74.90	—	4253	75.90	—
RTE	NORM	123	74.10	—	1408	76.51	—
MRPC	UNREG	17	84.74	89.04	2176	85.42	89.29
MRPC	UNREG	42	85.01	89.26	512	85.15	89.11
MRPC	UNREG	123	85.01	89.07	2304	85.42	89.42
MRPC	MIX	17	84.47	88.95	2760	84.47	88.95
MRPC	MIX	42	84.06	88.45	1280	85.29	89.07
MRPC	MIX	123	85.01	89.19	2070	85.69	89.51
MRPC	NORM	17	84.88	89.06	1792	85.83	89.60
MRPC	NORM	42	84.88	89.06	640	85.29	89.18
MRPC	NORM	123	84.60	88.78	1152	85.56	89.65

Calibration and confirmation rows are stored separately. Table 10 reports all fixed and validation-selected confirmation outcomes on the inner-selection split. The original-backbone probe reference, masked-token cross-entropy 2.0956, uses the frozen original language-model head on a fixed 256-example set. The practical adapter’s inserted state is recorded separately.

B.9 PAIRED CONFIRMATION CONTRASTS

Table 11 gives each seed difference, its mean and sample SD, and a nominal 95% paired Student- t interval ($n = 3$, two degrees of freedom). The intervals assume independent, approximately normal seed differences and describe variation on the fixed inner-selection split. They are unadjusted for

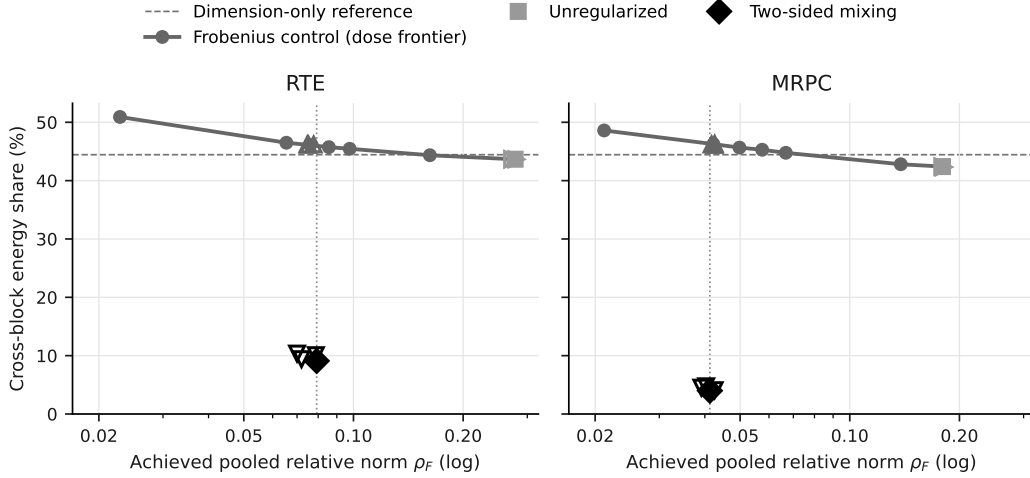


Figure 3: Interaction share across the tested size-penalty sweep. Filled circles show NORM calibration doses, squares UNREG and diamonds MIX. Open triangles show individual confirmation runs. The horizontal line is the 4/9 dimension-only reference; the vertical line marks the MIX calibration size. Shrinking the NORM update leaves a large interaction share.

Table 11: Paired fixed-endpoint differences (percentage points), left arm minus right. All three pairwise contrasts are reported for cross-block share and accuracy.

Task	Contrast	Metric	Seed 17	Seed 42	Seed 123	Mean $\pm$ SD	Nominal 95% CI
RTE	MIX-NORM	Cross	-36.80	-36.11	-35.78	-36.23 $\pm$ 0.52	[-37.51, -34.95]
RTE	MIX-NORM	Accuracy	1.00	-3.01	-4.22	-2.07 $\pm$ 2.73	[-8.87, 4.72]
RTE	MIX-UNREG	Cross	-34.24	-33.52	-33.33	-33.70 $\pm$ 0.48	[-34.89, -32.51]
RTE	MIX-UNREG	Accuracy	4.22	-1.00	-2.61	0.20 $\pm$ 3.57	[-8.67, 9.07]
RTE	NORM-UNREG	Cross	2.55	2.59	2.45	2.53 $\pm$ 0.07	[2.35, 2.71]
RTE	NORM-UNREG	Accuracy	3.21	2.01	1.61	2.28 $\pm$ 0.84	[0.20, 4.35]
MRPC	MIX-NORM	Cross	-41.55	-41.68	-42.02	-41.75 $\pm$ 0.25	[-42.36, -41.14]
MRPC	MIX-NORM	Accuracy	-0.41	-0.82	0.41	-0.27 $\pm$ 0.62	[-1.82, 1.28]
MRPC	MIX-UNREG	Cross	-37.66	-37.80	-38.24	-37.90 $\pm$ 0.30	[-38.65, -37.15]
MRPC	MIX-UNREG	Accuracy	-0.27	-0.95	0.00	-0.41 $\pm$ 0.49	[-1.63, 0.81]
MRPC	NORM-UNREG	Cross	3.89	3.88	3.79	3.85 $\pm$ 0.06	[3.71, 3.99]
MRPC	NORM-UNREG	Accuracy	0.14	-0.14	-0.41	-0.14 $\pm$ 0.27	[-0.81, 0.54]

multiple contrasts. For MIX-NORM accuracy, interval half-widths are 6.79 points on RTE and 1.55 on MRPC. Appendix B.18 separately calculates hypothetical power-based effect thresholds. Training seeds provide replication; modules and checkpoints describe each run.

B.10 MODULE NORMS UNDER POOLED MATCHING

For module ℓ , let $r_{a,\ell} = \|\Delta_{a,\ell}\|_F / \|W_{\text{pre},\ell}\|_F$. Table 12 compares the same 48 modules within each seed. Its rescaled check compares $(r_{\text{NORM},\ell} / \rho_{\text{NORM}}) / (r_{\text{MIX},\ell} / \rho_{\text{MIX}})$ with one, removing the residual global norm ratio descriptively without changing or retraining either arm. At most two modules per pair satisfy a 5% tolerance after this rescaling. Figure 4 shows where norms differ: RTE MIX concentrates its largest relative norms in output projections, notably layers 7 and 11, while many other modules have very small updates. The 10^{-6} threshold is a descriptive reporting threshold, not a training gate. Ratios against these near-zero updates can be arbitrarily large, so we show module norms directly. Absolute squared norms independently quantify energy concentration (Figure 5). For MRPC, the three largest modules carry 35.7–50.1% of MIX total energy versus 17.5–19.5% for NORM. These descriptive order statistics are accompanied by the complete cumulative curves; module counts are not additional experimental replicates.

Table 12: Signed pooled error $\delta_s = \rho_{\text{NORM},s}/\rho_{\text{MIX},s} - 1$ and module-relative norms. The two within-5% columns count matched module norms before and after global rescaling. The last column counts module norms below 10^{-6} for MIX/NORM. Modules describe heterogeneity within each seed.

Task	Seed	δ_s (%)	Median MIX	Median NORM	Within 5%	Rescaled	Small MIX/NORM
RTE	17	+8.0	0.0228	0.0788	1/48	0/48	14/0
RTE	42	-5.2	0.0209	0.0764	0/48	0/48	15/0
RTE	123	+7.5	0.0126	0.0746	0/48	0/48	12/0
MRPC	17	+5.5	0.0234	0.0430	4/48	2/48	0/0
MRPC	42	+6.7	0.0277	0.0426	1/48	1/48	1/0
MRPC	123	-0.1	0.0273	0.0420	2/48	2/48	1/0

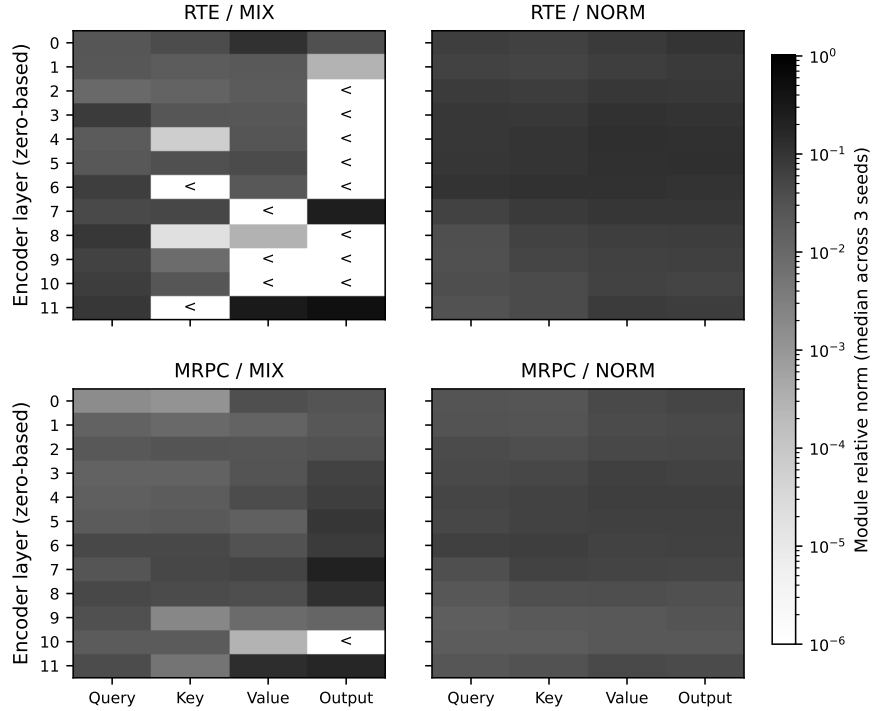


Figure 4: Module-relative update norms, median across three confirmation seeds for each module. All panels use the same logarithmic scale; “<” marks medians below the displayed 10^{-6} floor. This heatmap displays medians only; it does not show per-module seed dispersion. No zero-update energy fractions are inferred from these norms.

B.11 LEARNED PLACEMENT AFTER SUBTRACTING INSERTION

We evaluate $\Delta_{\text{learned}} = W_{\text{eff}}(T) - W_{\text{eff}}(0)$ directly, before squaring or normalizing. Subtracting initial energy from final energy would omit the cross term and is not this calculation. Every row is tied to the same fixed checkpoint and validation hash as the total-update report. The exported absolute energies of all 864 module records (18 runs, 48 modules each) independently reproduce pooled fractions, equal-module fractions and pooled relative norms to within 10^{-12} .

Table 13 distinguishes total and learned allocation. After subtracting insertion, MIX’s pooled leading share changes from 69.8% to 65.5% on MRPC and stays at 45.2% on RTE. Equal-module learned leading shares are 67.3% and 64.7%, respectively. The task contrast therefore depends on whether modules are weighted by energy or equally. Initialization matters most for small final updates: the largest per-seed change in an equal-module block fraction is 6.6 points on RTE and 13.4 on MRPC. Learned displacement includes any cancellation of the inserted contribution.

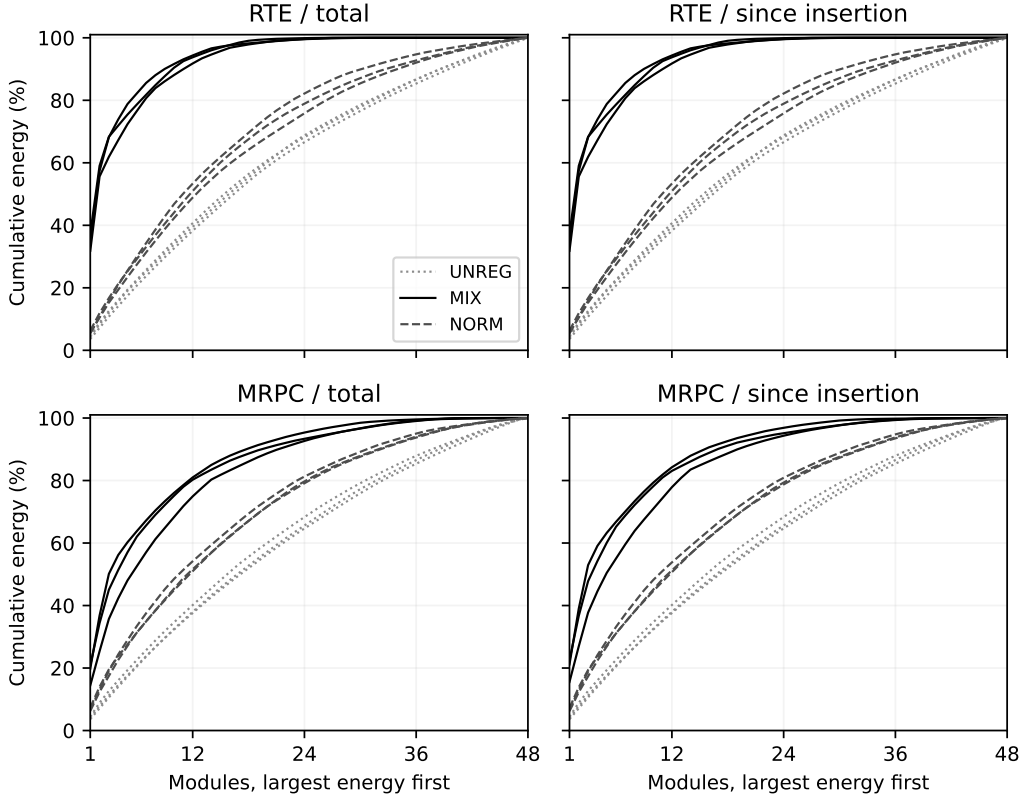


Figure 5: Cumulative absolute squared update norm across all 48 modules, sorted separately within each run. Each curve is one confirmation seed. The left panels include insertion; the right panels subtract it before computing energies. Steeper early accumulation indicates more concentrated adaptation. Sorting describes concentration, not correspondence of module identities across arms.

Table 13: Total and learned-since-insertion energy percentages at fixed endpoints, mean $\pm$ sample SD over three seeds. Cross combines LT and TL. Both normalizations use the same module energies; pooled weighting emphasizes modules with larger updates.

Task	Arm	Delta	Pooled energy			Equal module		
			LL	Cross	TT	LL	Cross	TT
RTE	UNREG	Total	45.9 $\pm$ 0.1	43.7 $\pm$ 0.0	10.4 $\pm$ 0.0	46.2 $\pm$ 0.1	43.5 $\pm$ 0.0	10.3 $\pm$ 0.1
RTE	UNREG	Learned	45.9 $\pm$ 0.1	43.7 $\pm$ 0.0	10.4 $\pm$ 0.0	46.2 $\pm$ 0.1	43.5 $\pm$ 0.0	10.3 $\pm$ 0.1
RTE	MIX	Total	45.2 $\pm$ 0.9	10.0 $\pm$ 0.5	44.8 $\pm$ 1.4	61.5 $\pm$ 2.1	14.9 $\pm$ 0.9	23.6 $\pm$ 1.8
RTE	MIX	Learned	45.2 $\pm$ 0.9	10.0 $\pm$ 0.5	44.8 $\pm$ 1.4	67.3 $\pm$ 3.0	10.0 $\pm$ 1.4	22.6 $\pm$ 2.0
RTE	NORM	Total	40.7 $\pm$ 0.0	46.2 $\pm$ 0.0	13.1 $\pm$ 0.0	40.5 $\pm$ 0.2	46.3 $\pm$ 0.1	13.2 $\pm$ 0.1
RTE	NORM	Learned	40.7 $\pm$ 0.0	46.2 $\pm$ 0.0	13.1 $\pm$ 0.0	40.5 $\pm$ 0.2	46.3 $\pm$ 0.1	13.2 $\pm$ 0.1
MRPC	UNREG	Total	48.0 $\pm$ 0.1	42.4 $\pm$ 0.1	9.6 $\pm$ 0.1	48.2 $\pm$ 0.0	42.2 $\pm$ 0.0	9.6 $\pm$ 0.0
MRPC	UNREG	Learned	47.9 $\pm$ 0.1	42.5 $\pm$ 0.0	9.7 $\pm$ 0.1	48.1 $\pm$ 0.0	42.4 $\pm$ 0.0	9.6 $\pm$ 0.0
MRPC	MIX	Total	69.8 $\pm$ 1.1	4.5 $\pm$ 0.3	25.8 $\pm$ 1.4	76.2 $\pm$ 0.5	5.5 $\pm$ 0.1	18.3 $\pm$ 0.6
MRPC	MIX	Learned	65.5 $\pm$ 1.1	5.1 $\pm$ 0.4	29.4 $\pm$ 1.5	64.7 $\pm$ 2.7	7.7 $\pm$ 0.8	27.6 $\pm$ 2.7
MRPC	NORM	Total	40.6 $\pm$ 0.2	46.2 $\pm$ 0.1	13.2 $\pm$ 0.1	40.5 $\pm$ 0.3	46.2 $\pm$ 0.1	13.3 $\pm$ 0.2
MRPC	NORM	Learned	41.3 $\pm$ 0.2	45.7 $\pm$ 0.1	13.0 $\pm$ 0.1	42.0 $\pm$ 0.2	45.1 $\pm$ 0.0	13.0 $\pm$ 0.2

B.12 EXECUTED FRESH-PROTOCOL SETTINGS AND INPUT VALIDATION

The fresh study uses the pinned RoBERTa-base checkpoint e2da8e2f811d1448a5b465c236feacd80ffbac7b and GLUE dataset revision bcdcb79d07bc864c1c254ccfcedcce55bcc9a8c. A fixed 20% inner-selection split is

Table 14: Every confirmation seed’s learned-since-insertion block percentages. Fixed endpoints and validation identities agree with Table 10. All four blocks are retained; rounding may affect displayed sums.

Task	Arm	Seed	LL	Pooled energy			LL	Equal module		
				LT	TL	TT		LT	TL	TT
RTE	UNREG	17	45.92	22.48	21.17	10.43	46.09	22.36	21.20	10.35
RTE	UNREG	42	46.02	22.25	21.37	10.36	46.26	22.17	21.32	10.25
RTE	UNREG	123	45.87	22.41	21.29	10.43	46.12	22.29	21.27	10.32
RTE	MIX	17	44.41	4.53	4.87	46.19	69.14	4.71	3.87	22.27
RTE	MIX	42	45.09	4.69	5.40	44.82	63.85	6.40	4.91	24.83
RTE	MIX	123	46.17	5.29	5.08	43.47	68.96	5.84	4.37	20.83
RTE	NORM	17	40.63	23.24	22.95	13.17	40.35	23.25	23.11	13.29
RTE	NORM	42	40.67	23.12	23.09	13.13	40.39	23.13	23.15	13.32
RTE	NORM	123	40.72	23.33	22.82	13.13	40.76	23.28	22.83	13.14
MRPC	UNREG	17	47.77	21.85	20.67	9.71	48.02	21.72	20.67	9.59
MRPC	UNREG	42	47.97	21.86	20.57	9.60	48.09	21.80	20.57	9.54
MRPC	UNREG	123	47.87	21.70	20.74	9.69	48.07	21.58	20.75	9.60
MRPC	MIX	17	66.12	2.66	2.74	28.47	63.19	3.99	4.60	28.21
MRPC	MIX	42	66.23	2.51	2.72	28.54	67.79	3.27	4.23	24.72
MRPC	MIX	123	64.22	2.23	2.43	31.12	63.01	3.06	3.92	30.01
MRPC	NORM	17	41.08	22.82	22.97	13.13	41.80	22.68	22.35	13.17
MRPC	NORM	42	41.35	22.76	22.89	12.99	42.01	22.44	22.64	12.91
MRPC	NORM	123	41.48	22.55	23.06	12.91	42.12	22.34	22.73	12.81

drawn from the training data with seed 271828; the official RTE and MRPC validation sets (277 and 408 examples) were held aside throughout training and selection, then evaluated once after the 18 checkpoints were frozen (Appendix B.15). Inspecting tokenization parity on held-aside input text is not a task-score evaluation. No model or checkpoint was selected using those held-aside labels.

Executed manifests specify float32 training, microbatch 8 with four accumulation steps (effective batch 32), maximum sequence length 128, gradient-norm clipping at 1, and zero weight decay. AdamW uses separate adapter and task-head groups with linear learning-rate decay to zero and no warmup. The adapter rate is 10^{-2} on both tasks; head rates are 5×10^{-4} on RTE and 10^{-3} on MRPC. Fixed endpoints are 5,670 and 2,760 optimizer steps, respectively. The practical adapter initializes both ambient scalars and tail coefficients at 0.01 for RTE and 0.1 for MRPC, with an identity leading block and no rotations. These initialization values belong to the practical study, not the strict-band branch. Task evaluation uses batches of 16; probe evaluation uses batches of 8. The secondary checkpoint maximizes inner-selection accuracy, excludes step 0, and breaks ties toward the earliest step. Candidates occur every 128 steps and at the registered fractions $\{0, .01, .05, .1, .25, .5, .75, 1\}$ of the budget. The fixed endpoint remains primary regardless of the selected score.

Before calibration, a 32-example, 200-step overfit gate required training accuracy at least 0.95; a 600-step unregularized learning gate required inner accuracy at least three points above the majority baseline. RTE passed at the adopted rates. On MRPC, the historical adapter rate 0.05 reached the majority baseline (0.6744) and failed the second gate; the adopted common rate 0.01 reached 0.8474. This change preceded the comparisons and applied to every fresh MRPC arm. It does not revise the historical results.

The tokenizer repair explicitly restores the BPE model type omitted from its upstream serialization. An independent byte-level BPE implementation exactly reproduced input IDs and attention masks for 181 examples across both tasks and all prepared splits, including 51 truncated examples. Both durable invalidation records are enforced during extraction: all 39 runs from the corrupted-input campaigns are excluded. Run manifests bind source files, inputs, recipes and validation reports; an exported validation hash identifies the report, while independent reaggregation here audits the export rather than rerunning training.

The frozen-head probe contains 256 sampled examples from WikiText-2-raw-v1’s test split (Merity et al., 2017), corpus revision b08601e04326c79dfdd32d625aee71d232d685c3. Sampling

Table 15: Identity alignment and residual leading structure (percentages, mean $\pm$ sample SD, three seeds). The first two columns weight modules by leading-block energy; the third weights modules equally. The final column uses all learned delta energy as denominator.

Task	Arm	Total $\eta_{I,\text{pool}}$	Learned $\eta_{I,\text{pool}}$	Learned $\eta_{I,\text{eqm}}$	Non-identity LL / learned total
RTE	UNREG	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	45.94 $\pm$ 0.08
RTE	MIX	84.67 $\pm$ 1.04	84.67 $\pm$ 1.05	79.67 $\pm$ 2.01	6.94 $\pm$ 0.60
RTE	NORM	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	40.67 $\pm$ 0.04
MRPC	UNREG	0.81 $\pm$ 0.07	0.29 $\pm$ 0.04	0.29 $\pm$ 0.05	47.73 $\pm$ 0.10
MRPC	MIX	91.38 $\pm$ 0.45	89.50 $\pm$ 0.61	79.23 $\pm$ 3.45	6.89 $\pm$ 0.51
MRPC	NORM	0.05 $\pm$ 0.01	2.76 $\pm$ 0.13	5.94 $\pm$ 0.37	40.16 $\pm$ 0.21

seed 271828 and masking seed 161803 fix the examples and 15% masking procedure. Summing per-example loss numerators and dividing by the 4,001 scored tokens gives original-backbone cross-entropy 2.095626. This reference uses CPU float32; adapted probes use GPU float32. The corpus is held out from task adaptation, but is not guaranteed unseen during pretraining. The probe measures behavior under fixed examples and masks, not preserved knowledge in general.

B.13 IDENTITY ALIGNMENT OF THE LEARNED LEADING BLOCK

For each module, project C_{LL} onto its own identity matrix as in (37). Table 15 reports the fraction of summed leading energy explained by these module-wise projections, $\eta_{I,\text{pool}} = 1 - \sum_{\ell} \|C_{LL,\ell} - (\text{tr } C_{LL,\ell}/k)I_k\|_F^2 / \sum_{\ell} \|C_{LL,\ell}\|_F^2$. This permits a different fitted scalar in each module; it does not fit one scalar to the whole network. We also report the equal-module mean of η_I and the non-identity leading residual divided by total delta energy. All quantities use exported checkpoint diagnostics, not a fit to the rounded block fractions.

MIX’s learned leading energy is mostly aligned with the identity component. Its non-identity leading residual is about 6.9% of total learned energy on both tasks. Learned cross shares are 10.0% and 5.1%, so the endpoint retains departures from the constant-scaler limit, which has exactly zero cross energy. Identity alignment describes the leading block; centered-scaler measurements and random-projector controls would distinguish the factor mechanisms that produce it.

B.14 SENSITIVITY TO MODULES WITH SMALL UPDATES

This descriptive sensitivity analysis was specified after inspecting the original outcomes. For each task and seed, retain the same named modules in MIX and NORM only if both total-update relative norms are at least the stated floor. Apply that identical subset to total and learned diagnostics, then average module cross fractions. The two floors, 10^{-6} and 10^{-3} , are reporting choices, not training gates or preregistered hypotheses. Counts and all seed differences are retained in the analysis data; the seed remains the unit of replication.

Table 16 shows that the gap survives both restrictions. We recompute each retained module’s fraction directly, including small nonzero updates. The restriction conditions on attained norms, a post-treatment variable. It is a descriptive check of averaging sensitivity; matching module energy during training would be a different intervention.

B.15 ONE-TIME EVALUATION ON HELD-ASIDE TASK SPLITS

After freezing all 18 fixed-endpoint confirmation checkpoints, we evaluated them once on the official RTE (277 examples) and MRPC (408 examples) validation sets. The frozen recipes, doses and checkpoints were retained throughout. CPU loading reproduced inner-selection accuracy and MRPC F1 exactly, with task loss agreeing within the recorded numerical tolerance. Exported per-example IDs, labels, predictions and losses reproduce the held-out metrics. Checkpoint hashes and validation-report identities link every row to the frozen population.

Held-out accuracies for UNREG/MIX/NORM are $71.72 \pm 1.50/71.36 \pm 1.63/71.24 \pm 1.63$ on RTE and $86.19 \pm 0.93/85.46 \pm 1.02/85.78 \pm 0.65$ on MRPC. MRPC F1 is $90.12 \pm 0.65/89.68 \pm$

Table 16: Equal-module cross shares and paired NORM-MIX differences (percentage points), mean $\pm$ sample SD over three seeds. Counts are retained modules in seed order 17, 42, 123; each seed starts with 48.

Task	Norm floor	Delta	Counts	MIX	NORM	NORM-MIX
RTE	$1e-06$	Total	34,33,36	14.68 ± 1.80	46.72 ± 0.13	32.04 ± 1.71
RTE	$1e-06$	Learned	34,33,36	14.06 ± 2.21	46.72 ± 0.13	32.66 ± 2.10
RTE	0.001	Total	30,32,30	15.19 ± 1.84	46.76 ± 0.18	31.57 ± 1.83
RTE	0.001	Learned	30,32,30	15.19 ± 1.86	46.76 ± 0.18	31.57 ± 1.84
MRPC	$1e-06$	Total	48,47,47	5.46 ± 0.25	46.24 ± 0.04	40.78 ± 0.21
MRPC	$1e-06$	Learned	48,47,47	7.79 ± 0.74	45.14 ± 0.09	37.35 ± 0.83
MRPC	0.001	Total	47,44,46	5.63 ± 0.38	46.26 ± 0.03	40.63 ± 0.38
MRPC	0.001	Learned	47,44,46	8.08 ± 0.75	45.23 ± 0.09	37.15 ± 0.84

Table 17: All frozen confirmation endpoints on the held-aside official splits. Accuracy and F1 are percentages; loss is task cross-entropy. These scores were not used for selection.

Task	Arm	Seed	Accuracy	F1	Loss
RTE	UNREG	17	70.04	—	2.347
RTE	UNREG	42	72.20	—	2.461
RTE	UNREG	123	72.92	—	2.478
RTE	MIX	17	69.68	—	1.330
RTE	MIX	42	71.48	—	1.192
RTE	MIX	123	72.92	—	1.165
RTE	NORM	17	69.68	—	2.567
RTE	NORM	42	72.92	—	1.956
RTE	NORM	123	71.12	—	2.057
MRPC	UNREG	17	85.78	89.75	0.872
MRPC	UNREG	42	85.54	89.74	0.938
MRPC	UNREG	123	87.25	90.88	0.802
MRPC	MIX	17	84.31	88.97	0.425
MRPC	MIX	42	85.78	89.97	0.370
MRPC	MIX	123	86.27	90.11	0.388
MRPC	NORM	17	86.27	90.24	0.719
MRPC	NORM	42	85.05	89.35	0.829
MRPC	NORM	123	86.03	90.02	0.718

0.62/89.87 $\pm$ 0.46. Table 17 gives every endpoint; Table 18 gives every paired contrast. Intervals use the three-seed assumptions in Appendix B.9 and are unadjusted for multiple contrasts.

Across the three confirmation seeds, MIX’s norm coefficients of variation are about 6.2% on RTE and 4.2% on MRPC. This variation is comparable to the calibration rule’s 5% tolerance. The confirmation report therefore retains every achieved error alongside the original doses and tolerance. Using several calibration seeds is a direct extension of this protocol.

B.16 HELD-ASIDE LOSS AND CONFIDENCE CALIBRATION

Task negative log likelihood (NLL) is the cross-entropy column in Table 17. UNREG/MIX/NORM mean NLL is 2.429/1.229/2.193 nats on RTE and 0.871/0.395/0.755 on MRPC. MIX has lower loss in every paired seed, and its three-seed range is below both comparator ranges within each task. Table 19 reports all contrasts. We identified this ordering after inspecting the completed evaluation.

NLL depends on the probabilities assigned to labels. Changing logit scale can change NLL while preserving the predicted class. Temperature scaling (Guo et al., 2017) tests this explanation by fitting a positive temperature on inner-selection logits, freezing it, and then measuring held-out loss and calibration. The required inner-selection logits are absent from the current export, so the

Table 18: Paired held-aside differences (percentage points), left minus right; all contrasts retained. Nominal 95% intervals are descriptive, not multiplicity-adjusted findings.

Task	Contrast	Metric	Seed 17	Seed 42	Seed 123	Mean $\pm$ SD	Nominal 95% CI
RTE	MIX-NORM	accuracy	0.00	-1.44	1.81	0.12 $\pm$ 1.63	[-3.92, 4.16]
RTE	MIX-UNREG	accuracy	-0.36	-0.72	0.00	-0.36 $\pm$ 0.36	[-1.26, 0.54]
RTE	NORM-UNREG	accuracy	-0.36	0.72	-1.81	-0.48 $\pm$ 1.27	[-3.63, 2.67]
MRPC	MIX-NORM	accuracy	-1.96	0.74	0.25	-0.33 $\pm$ 1.44	[-3.89, 3.24]
MRPC	MIX-NORM	f1	-1.28	0.61	0.09	-0.19 $\pm$ 0.98	[-2.62, 2.23]
MRPC	MIX-UNREG	accuracy	-1.47	0.25	-0.98	-0.74 $\pm$ 0.88	[-2.93, 1.46]
MRPC	MIX-UNREG	f1	-0.79	0.23	-0.77	-0.44 $\pm$ 0.58	[-1.89, 1.00]
MRPC	NORM-UNREG	accuracy	0.49	-0.49	-1.23	-0.41 $\pm$ 0.86	[-2.55, 1.73]
MRPC	NORM-UNREG	f1	0.49	-0.38	-0.86	-0.25 $\pm$ 0.69	[-1.95, 1.45]

Table 19: Paired held-aside task NLL differences in nats, left minus right. Sample SD uses the $n - 1$ divisor, $n = 3$. Intervals follow Appendix B.9; negative values favor the left arm on this loss only.

Task	Contrast	Seed 17	Seed 42	Seed 123	Mean $\pm$ SD	Nominal 95% CI
RTE	MIX-NORM	-1.236	-0.764	-0.892	-0.964 $\pm$ 0.244	[-1.571, -0.357]
RTE	MIX-UNREG	-1.017	-1.269	-1.313	-1.199 $\pm$ 0.160	[-1.596, -0.803]
RTE	NORM-UNREG	+0.220	-0.505	-0.421	-0.235 $\pm$ 0.396	[-1.219, 0.749]
MRPC	MIX-NORM	-0.293	-0.459	-0.330	-0.361 $\pm$ 0.087	[-0.578, -0.144]
MRPC	MIX-UNREG	-0.447	-0.568	-0.414	-0.476 $\pm$ 0.081	[-0.677, -0.276]
MRPC	NORM-UNREG	-0.154	-0.108	-0.084	-0.115 $\pm$ 0.035	[-0.203, -0.028]

reported values are the models’ original losses. This comparison tests the complete regularization intervention, including its effects on scale and update allocation.

B.17 DESCRIPTIVE SENSITIVITY ALONG THE MAGNITUDE FRONTIER

For each task, we fit $p_{\text{cross}}^{(\%)} = a + b \log \rho_F$ by ordinary least squares to all seven NORM calibration doses (seed 31415). Table 20 reports the slope, maximum fitted residual and illustrative change $|b| \log(1.08)$. The latter is about 0.22 points, compared with the observed 36–42 point confirmation contrast. At signed norm error δ_s , the corresponding fitted change is $b \log(1 + \delta_s)$; all six values are in the analysis export. This is an interpolation of one observed frontier per task. It describes sensitivity within the measured range rather than bounding an exactly matched or unobserved experiment.

B.18 POWER SENSITIVITY UNDER EXPLICIT VARIANCE ASSUMPTIONS

The nominal intervals describe uncertainty; their half-widths are not minimum detectable effects. To make the small-seed limitation quantitative, consider a hypothetical two-sided paired t test with $n = 3$, independent Gaussian seed differences, significance level 0.05, and fixed true difference SD σ . Its power at a mean difference δ is

$$\Pr\{|T_{2, \sqrt{3}\delta/\sigma}| > t_{0.975, 2}\}. \quad (41)$$

Here $T_{\nu, \lambda}$ is noncentral Student t . Solving for 80% power gives $|\delta| \simeq 3.26\sigma$. Table 21 substitutes each observed paired sample SD as an uncertain variance scenario, including all task-score, task-NLL and probe contrasts. These are post hoc design sensitivities, not achieved power, effects excluded by the experiment, or an equivalence margin. At three seeds the variance itself is poorly estimated; all entries scale linearly with the assumed σ and condition on the fixed evaluation examples.

B.19 STRICTLY CONFINED ADAPTATION: COMPLETE RTE RESULTS

Training and evaluation protocol. We use the pretrained RoBERTa-base encoder and a newly initialized two-layer classification head that maps its sentence-pair representation to two label scores. Softmax converts these scores to probabilities. All conditions train this head; only adapter conditions

Table 20: Single-calibration-seed NORM frontier. Cross shares and fitted changes are percentage points. The norm range is computed within each task, not across tasks. The illustrative shift is a fitted sensitivity, not a causal bound.

Task	Norm max/min	Cross range	b (per log norm)	8% shift	Max residual
RTE	12.10	43.66–50.91	-2.905	0.224	0.878
MRPC	8.47	42.41–48.60	-2.900	0.223	0.301

Table 21: Effect magnitude for 80% power in the stated hypothetical Gaussian paired test, using the observed SD as a variance scenario. This is a planning calculation, not a new test of the reported contrasts. pp: percentage points.

Task	Contrast	Metric	Paired SD	80%-power effect
RTE	MIX–NORM	Accuracy (pp)	1.628	5.313
RTE	MIX–NORM	Task NLL (nat)	0.244	0.798
RTE	MIX–NORM	Probe CE (nat)	0.262	0.854
RTE	MIX–UNREG	Accuracy (pp)	0.361	1.178
RTE	MIX–UNREG	Task NLL (nat)	0.160	0.521
RTE	MIX–UNREG	Probe CE (nat)	0.313	1.021
RTE	NORM–UNREG	Accuracy (pp)	1.268	4.138
RTE	NORM–UNREG	Task NLL (nat)	0.396	1.293
RTE	NORM–UNREG	Probe CE (nat)	0.099	0.324
MRPC	MIX–NORM	Accuracy (pp)	1.436	4.688
MRPC	MIX–NORM	F1 (pp)	0.976	3.185
MRPC	MIX–NORM	Task NLL (nat)	0.087	0.285
MRPC	MIX–NORM	Probe CE (nat)	0.249	0.812
MRPC	MIX–UNREG	Accuracy (pp)	0.884	2.884
MRPC	MIX–UNREG	F1 (pp)	0.581	1.895
MRPC	MIX–UNREG	Task NLL (nat)	0.081	0.264
MRPC	MIX–UNREG	Probe CE (nat)	0.234	0.764
MRPC	NORM–UNREG	Accuracy (pp)	0.861	2.810
MRPC	NORM–UNREG	F1 (pp)	0.685	2.237
MRPC	NORM–UNREG	Task NLL (nat)	0.035	0.115
MRPC	NORM–UNREG	Probe CE (nat)	0.113	0.369

also train coefficients and, where enabled, rotations. The original encoder weights remain frozen. A fixed 20% split of RTE’s training set, drawn with seed 271828, supplies the inner-selection examples; training uses the remaining 80%. Conditions share the head initialization and data order within each seed. All train for 5,670 AdamW steps with effective batch size 32, adapter learning rate 10^{-2} , head learning rate 5×10^{-4} , linear decay, no warmup, no weight decay, and no interaction penalty. Other input and optimizer settings follow Appendix B.12; the zero-update initialization below replaces the practical adapter’s initialization. We report accuracy and cross-entropy on the inner-selection split at the fixed final step, not the best-scoring checkpoint. The official RTE validation set has not been evaluated for this study.

The separate RTE study crosses three 256-direction bands with diagonal coefficients or rotations in the last 64 directions, plus an original-backbone head-only reference. Leading, middle and tail bands are [0, 256), [256, 512) and [512, 768). All 48 attention updates use $\Delta = U_B H V_B^\top$, start at zero, and omit the ambient scalars and leading-identity contribution. Three seeds (17, 42, 123) give 21 registered confirmations. Two seed-31415 timing pilots are recorded separately. Off-band energy is structurally zero, with a 10^{-8} float32 layer-test bound.

Each rotation uses PyTorch 2.5.1’s `nn.utils.parametrizations.orthogonal` with `orthogonal_map="matrix_exp"` and the default `use_trivialization=True`. Its square weight is initialized to the identity before registration. The matrix exponential maps a skew-symmetric parameter matrix to an orthogonal output throughout optimization. Each pair stores $2q^2$

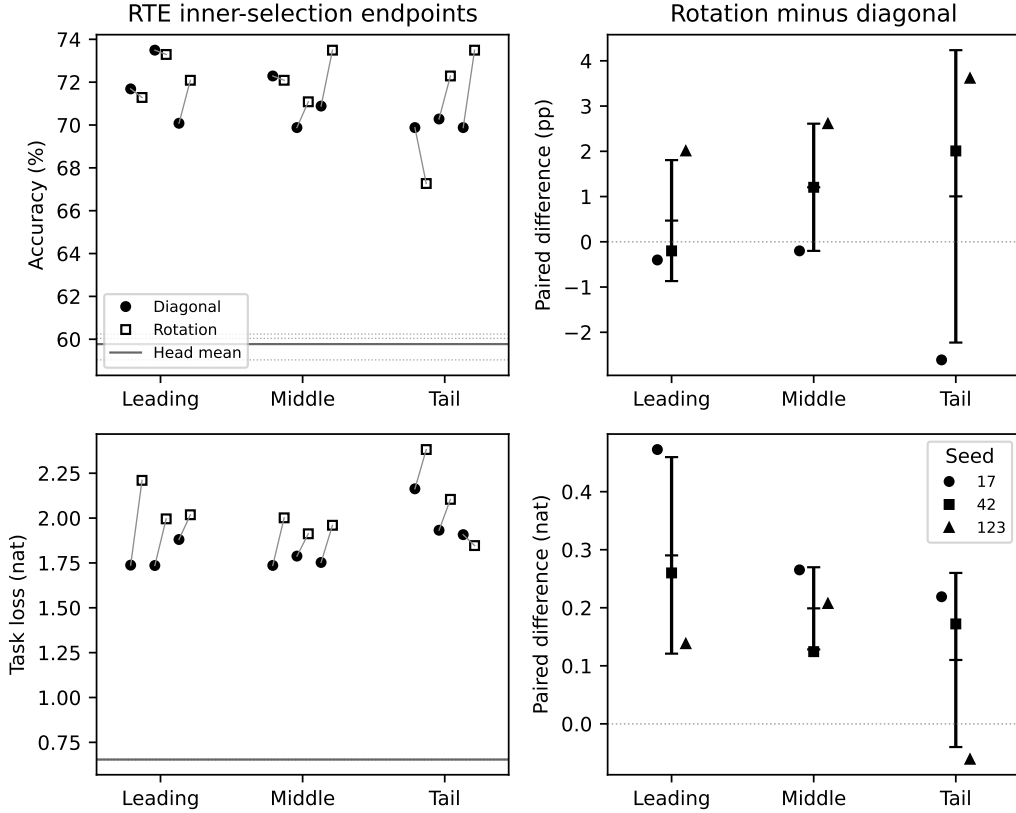


Figure 6: All three seeds for each band and flexibility level, at the fixed inner-selection endpoint. Left: diagonal and rotated outcomes are joined within seed; horizontal lines show head-only references. Right: every paired rotation-minus-diagonal difference, with mean and sample SD. Rotations create additional within-band interaction without a consistent accuracy improvement; the loss panels show the accompanying confidence-sensitive behavior.

trainable entries, while its two orthogonal outputs have $q(q-1)$ combined degrees of freedom (Appendix C).

The final export contains all 21 confirmation outcomes and the two timing pilots. Table 22 summarizes every arm; Table 23 reports all 18 registered paired contrasts; Table 24 gives every individual endpoint. All scores are inner-selection results. The freeze ledger joins all 21 fixed checkpoints to their validation hashes and the same registered protocol. Although repository revisions differ across launches, their experiment-source trees are identical; the supplement verifies all 46 recorded source-file fingerprints against a saved source snapshot. Within-band off-diagonal energy is computed directly from compact cores, with signed floating-point residuals retained in the raw export for diagonal arms.

The tail-capacity contrast compares tail-only and head-only training at the same fixed endpoint, with common initial heads and data order. The flexibility contrast compares rotated and diagonal adapters at the same band width and training budget. Leading and middle bands provide location controls. Row order and these contrasts were fixed before confirmation inspection, after disclosure of the tail pilot. The later emphasis on the tail question leaves the full registered population intact.

Rotated updates place 11.36%, 10.16% and 7.88% of their energy off the diagonal within the leading, middle and tail bands, respectively. The rotations therefore change the trained core in every band. Mean accuracy changes relative to diagonal adapters are +0.47, +1.20 and +1.00 points; all paired intervals include zero. Mean prediction loss increases by 0.290, 0.199 and 0.110 nats. The diagonal tail’s accuracy gain over head-only training occurs in all three seeds, together with a mean

Table 22: Complete RTE band comparison, mean $\pm$ sample SD, three seeds per row. Accuracy and loss use the inner-selection split. Off-diagonal energy is a percentage within the selected band, computed directly from the compact trained cores; diagonal residuals are float32 roundoff. The head-only backbone update is zero, so its energy fraction is undefined. All nonzero updates have structural off-band zero.

Condition	Accuracy (%)	Task loss (nat)	Within-band off-diag. (%)	ρ_F
Head only	59.77 $\pm$ 0.65	0.654 $\pm$ 0.002	—	0
Leading / diagonal	71.75 $\pm$ 1.71	1.785 $\pm$ 0.083	0.00 $\pm$ 0.00	0.3283 $\pm$ 0.0009
Leading / rotation	72.22 $\pm$ 1.01	2.075 $\pm$ 0.118	11.36 $\pm$ 0.41	0.3090 $\pm$ 0.0006
Middle / diagonal	71.02 $\pm$ 1.21	1.759 $\pm$ 0.027	0.00 $\pm$ 0.00	0.3287 $\pm$ 0.0025
Middle / rotation	72.22 $\pm$ 1.21	1.958 $\pm$ 0.044	10.16 $\pm$ 0.29	0.3201 $\pm$ 0.0030
Tail / diagonal	70.01 $\pm$ 0.23	2.001 $\pm$ 0.141	0.00 $\pm$ 0.00	0.2560 $\pm$ 0.0017
Tail / rotation	71.02 $\pm$ 3.30	2.111 $\pm$ 0.268	7.88 $\pm$ 0.31	0.2720 $\pm$ 0.0085

Table 23: All registered paired RTE contrasts at the fixed endpoint on inner-selection examples. Entries are mean $\pm$ sample SD of the three seed differences and nominal 95% paired t intervals (two degrees of freedom). Rotation-gain differences compare rotation-minus-diagonal changes between bands. Intervals are descriptive and unadjusted for multiple comparisons.

Contrast	Accuracy (pp)	Interval	Task loss (nat)	Interval
Leading / diagonal – head	11.98 $\pm$ 1.71	[7.74, 16.22]	1.130 $\pm$ 0.085	[0.920, 1.341]
Leading / rotation – head	12.45 $\pm$ 0.53	[11.13, 13.77]	1.421 $\pm$ 0.117	[1.131, 1.711]
Middle / diagonal – head	11.24 $\pm$ 1.84	[6.67, 15.82]	1.105 $\pm$ 0.027	[1.038, 1.171]
Middle / rotation – head	12.45 $\pm$ 1.41	[8.96, 15.94]	1.304 $\pm$ 0.044	[1.194, 1.413]
Tail / diagonal – head	10.24 $\pm$ 0.53	[8.92, 11.56]	1.347 $\pm$ 0.140	[1.000, 1.693]
Tail / rotation – head	11.24 $\pm$ 2.70	[4.53, 17.96]	1.457 $\pm$ 0.266	[0.796, 2.118]
LEAD rotation – diagonal	0.47 $\pm$ 1.34	[−2.85, 3.79]	0.290 $\pm$ 0.169	[−0.130, 0.711]
MID rotation – diagonal	1.20 $\pm$ 1.41	[−2.29, 4.70]	0.199 $\pm$ 0.071	[0.023, 0.375]
TAIL rotation – diagonal	1.00 $\pm$ 3.23	[−7.02, 9.03]	0.110 $\pm$ 0.150	[−0.263, 0.483]
MID – LEAD / diagonal	−0.74 $\pm$ 2.49	[−6.93, 5.46]	−0.026 $\pm$ 0.093	[−0.256, 0.204]
MID – LEAD / rotation	−0.00 $\pm$ 1.94	[−4.81, 4.81]	−0.117 $\pm$ 0.081	[−0.317, 0.083]
Rotation gain: MID – LEAD	0.74 $\pm$ 0.61	[−0.79, 2.26]	−0.091 $\pm$ 0.144	[−0.448, 0.265]
TAIL – LEAD / diagonal	−1.74 $\pm$ 1.51	[−5.48, 2.00]	0.216 $\pm$ 0.200	[−0.280, 0.713]
TAIL – LEAD / rotation	−1.20 $\pm$ 2.72	[−7.95, 5.54]	0.036 $\pm$ 0.183	[−0.418, 0.491]
Rotation gain: TAIL – LEAD	0.54 $\pm$ 2.40	[−5.42, 6.49]	−0.180 $\pm$ 0.084	[−0.390, 0.030]
TAIL – MID / diagonal	−1.00 $\pm$ 1.41	[−4.50, 2.49]	0.242 $\pm$ 0.160	[−0.156, 0.640]
TAIL – MID / rotation	−1.20 $\pm$ 3.19	[−9.12, 6.71]	0.153 $\pm$ 0.249	[−0.466, 0.772]
Rotation gain: TAIL – MID	−0.20 $\pm$ 1.92	[−4.96, 4.56]	−0.089 $\pm$ 0.162	[−0.492, 0.314]

loss increase of 1.347 nats. These results separate the ability to learn useful classifications from improvements in the probabilities assigned to them.

The export contains endpoints rather than complete training curves, so we make no convergence-rate claim. A zero-initialized core gives zero rotation gradients at insertion; the final nonzero within-band off-diagonal energy shows that training leaves the diagonal family. Rotations add trainable parameters, so the comparison holds spectral support and training budget fixed while varying flexibility.

B.20 ARCHIVED BLOCK ALLOCATION AND FACTOR ASYMMETRY

The aggregate off-tail ratio combines direct leading adaptation and cross interaction. To separate them, we reconstruct the four energy fractions from the saved per-matrix diagnostics, square before averaging, and retain equal weighting of modules within each run. Figure 7 shows the resulting leading, cross, and tail allocation. Appendix B.21 gives all four blocks and accounts for the small finite-precision difference between the recorded coordinate and ambient norms.

Without regularization, cross blocks contain approximately 43% of update energy, the leading block 46%, and the tail 10%. Under C, cross share is 10–27%, tail share 22–34%, and leading share 44–64%; the latter rises in four task means. The retained leading block agrees with the structure

Table 24: All 21 strict-band/head confirmations and two timing pilots. Scores use fixed endpoints on the inner-selection split; each row is one run. Pilots are excluded from every confirmation summary. Adapter counts exclude the common 592,130-parameter trained head. Band updates remain inside their selected pretrained span by construction.

Task	Band	Core	Seed	Acc. (%)	Task loss	Adapter params
<i>All 21 confirmation endpoints</i>						
RTE	Lead	Diagonal	17	71.69	1.738	12,288
RTE	Lead	Diagonal	42	73.49	1.736	12,288
RTE	Lead	Diagonal	123	70.08	1.881	12,288
RTE	Lead	Rotated 64	17	71.29	2.210	405,504
RTE	Lead	Rotated 64	42	73.29	1.996	405,504
RTE	Lead	Rotated 64	123	72.09	2.019	405,504
RTE	Mid	Diagonal	17	72.29	1.736	12,288
RTE	Mid	Diagonal	42	69.88	1.788	12,288
RTE	Mid	Diagonal	123	70.88	1.753	12,288
RTE	Mid	Rotated 64	17	72.09	2.001	405,504
RTE	Mid	Rotated 64	42	71.08	1.913	405,504
RTE	Mid	Rotated 64	123	73.49	1.960	405,504
RTE	Tail	Diagonal	17	69.88	2.163	12,288
RTE	Tail	Diagonal	42	70.28	1.932	12,288
RTE	Tail	Diagonal	123	69.88	1.908	12,288
RTE	Tail	Rotated 64	17	67.27	2.382	405,504
RTE	Tail	Rotated 64	42	72.29	2.104	405,504
RTE	Tail	Rotated 64	123	73.49	1.847	405,504
RTE	Head only	—	17	59.04	0.656	0
RTE	Head only	—	42	60.24	0.655	0
RTE	Head only	—	123	60.04	0.652	0
<i>Timing pilots (excluded from confirmations)</i>						
RTE	Tail	Diagonal	31415	70.28	1.884	12,288
RTE	Tail	Rotated 64	31415	70.48	2.104	405,504

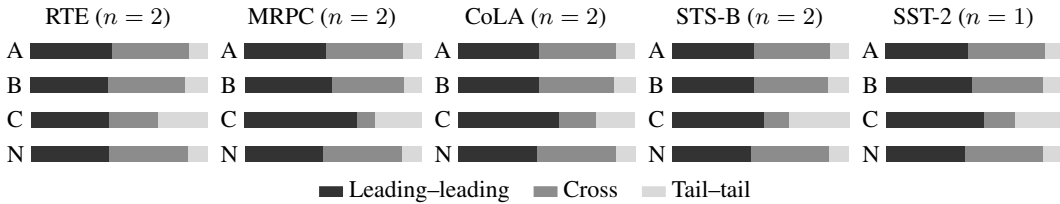


Figure 7: Leading/cross/tail allocation, averaged over modules then n seeds. A: no penalty; B: left-only; C: two-sided. N is the dimension-only reference (4/9, 4/9, 1/9), not a training run. C reduces cross share but retains a leading block; Table 6 supplies magnitude and dispersion, and Appendix B.21 supplies each seed’s four fractions.

in Proposition 2. These are fractions of the total update, which includes insertion. Appendix B.3 calculates the prescribed initial contribution, and the fresh study separately measures learned displacement and leading identity alignment.

Appendix B.21 gives each run’s mean block fractions; the supplementary archive contains all module records.

One-sided control and factor asymmetry. B has the highest mean score on RTE, MRPC and CoLA while its cross shares remain about 40–44% (Table 6). The score ordering therefore differs from the interaction ordering. In all nine A/B pairs, μ_E falls and ν_D rises. Reciprocal rescaling of the two factors can also produce this asymmetry, as shown in Appendix A.9; the effective update is needed to interpret the factor measurements.

Interpreting normalized geometry. The archived intervention changes normalized block allocation. Uniformly rescaling a fixed trained matrix would preserve those fractions. A size penalty during training acts through the adapter factors and can change them, which motivates the separate controlled experiment in Section 5.2. The archived measurements describe the selected endpoints.

An isotropically oriented perturbation has expected leading/cross/tail fractions k^2/d^2 , $2kr/d^2$, and r^2/d^2 (Appendix A.6). Figure 7 makes this dimension-only reference explicit. A lies nearby, but agreement of three aggregate fractions is not a test of isotropy or evidence that pretrained structure is unused. Equal-dimensional random-frame controls and measurements of other adapters are needed to distinguish those interpretations.

B.21 RECONSTRUCTING FOUR-BLOCK ALLOCATION

The 27 run summaries are accompanied by 1,296 module-level records: 48 adapted matrices for each A/B/C run. Their saved Frobenius leakage ratios use $\|\Delta\|_F$ in the denominator, whereas the off-tail ratio uses $\|C\|_F$. Write these recorded quantities as ℓ_{ab} for $ab \in \{LL, LT, TL\}$ and q for the off-tail ratio. To obtain an allocation normalized in the recorded coordinate frame, we compute per matrix

$$\hat{p}_{ab} = q^2 \frac{\ell_{ab}^2}{\ell_{LL}^2 + \ell_{LT}^2 + \ell_{TL}^2}, \quad \hat{p}_{TT} = 1 - q^2. \quad (42)$$

For exact orthogonal bases these agree with the fractions defined in Section 4. With saved finite-precision bases, the inferred discrepancy $|(\ell_{LL}^2 + \ell_{LT}^2 + \ell_{TL}^2)/q^2 - 1|$ is at most $2.64 \cdot 10^{-4}$ across the archive. The normalization makes the four recorded fractions sum to one without assuming perfect numerical orthogonality. Diagnostic denominators also include a 10^{-12} numerical stabilizer. We square each matrix’s ratios before averaging, never a task-mean ratio.

The analysis script reconciles every shared summary metric with the mean of its 48 saved module values before calculating these fractions. It excludes smoke tests and unmatched C-only reruns. Cross share falls in 410 of 432 matched A/C module pairs, q_{off} in 332, and left/right drift in 260/256; both drift run means fall in all nine pairs. These are within-run heterogeneity counts, not independent replication or uniform layerwise stabilization.

B.22 A REUSABLE SPECTRAL REPORT

Use a fixed pretrained checkpoint, adapted module list, and declared cutoff for all methods. The primary cutoff here follows the adapter tail dimension, not a measured spectral gap. For each saved effective matrix, record both $\Delta_{\text{total}} = W_{\text{eff}} - W_0$ and $\Delta_{\text{learned}} = W_{\text{eff}} - W_{\text{eff}}(0)$; report zero-update fractions as undefined. The report comprises absolute Frobenius block energies, their fractions, ρ_F , operator norm, stable rank, and the dimension-only fractions. Add leading identity alignment η_I and absolute non-identity residual energy under both delta definitions. Export each module’s norm and absolute block energies, the cumulative energy curve, and threshold-count curves; a cumulative energy curve alone cannot recover relative-norm thresholds. We report 10^{-6} for small-norm counts and both 10^{-6} , 10^{-3} for sensitivity subsets. Future comparisons must declare thresholds before inspection and retain the underlying values. Use the aggregation specified by each table: the focused headline pools squared energies, while equal-module summaries normalize before averaging. This report is diagnostic, not a preservation pass/fail criterion.

Let P, P' be rank- k left projectors before and after adaptation, and $m = \min(k, d - k)$. Supplement the largest-angle sine with

$$d_{\text{ch}} = \frac{\|P - P'\|_F}{\sqrt{2m}}, \quad o = \frac{\text{tr}(PP')}{k}, \quad d_{\text{ch}}^2 = \frac{k}{m}(1 - o). \quad (43)$$

Here d_{ch}^2 averages squared sines over the m potentially nonzero angles, including any zeros within them; o is normalized overlap, with random-independent-subspace expectation k/d . Apply the same definitions on the right. Report the boundary values σ_k, σ_{k+1} , their gap, and a prespecified cutoff sweep, such as $k \in \{16, 64, 128, 256, 512\}$. Singular-subspace perturbation bounds require separation conditions (Wedin, 1972); a norm/gap quotient is not an unconditional prediction of drift. A boundary tie is flagged, not presented as uniquely defined ranked stability.

Table 25: Four-block energy percentages for every archived seed, averaged over its 48 modules. Averaging seeds gives Figure 7. The middle columns sum to cross share; rows sum to 100% up to rounding. SST-2 has only seed 42.

Task	Condition	Seed	Leading-leading	Leading-tail	Tail-leading	Tail-tail
RTE	A	17	46.61	22.18	21.11	10.10
RTE	A	42	46.17	22.05	21.47	10.31
RTE	B	17	43.37	19.86	23.99	12.77
RTE	B	42	43.89	20.08	23.18	12.86
RTE	C	17	43.45	13.09	13.00	30.46
RTE	C	42	45.16	14.72	14.10	26.02
MRPC	A	17	46.41	22.20	21.15	10.24
MRPC	A	42	46.72	22.10	21.04	10.13
MRPC	B	17	46.81	21.77	20.05	11.37
MRPC	B	42	52.45	23.99	14.82	8.74
MRPC	C	17	62.62	6.77	5.77	24.85
MRPC	C	42	64.68	3.89	3.74	27.69
CoLA	A	17	45.89	22.58	21.10	10.43
CoLA	A	42	45.59	22.39	21.43	10.59
CoLA	B	17	45.25	21.18	21.37	12.20
CoLA	B	42	45.98	21.54	20.69	11.80
CoLA	C	17	57.57	9.96	7.15	25.33
CoLA	C	42	56.39	14.78	10.29	18.54
STS-B	A	17	46.24	22.32	21.15	10.29
STS-B	A	42	46.28	22.27	21.18	10.27
STS-B	B	17	45.48	20.35	21.78	12.39
STS-B	B	42	47.21	21.90	19.90	10.99
STS-B	C	17	52.98	8.74	5.93	32.35
STS-B	C	42	51.55	7.75	5.01	35.69
SST-2	A	42	46.52	22.38	20.95	10.15
SST-2	B	42	48.76	21.51	18.68	11.05
SST-2	C	42	55.74	9.26	7.94	27.06

Pair these measurements with validation and untouched-test performance, and one frozen-pretraining-head behavior probe on the same checkpoints when retention is studied. Store per-seed values at both a fixed training step and the selected checkpoint; log diagnostics separately from optimization costs. Full ranked-subspace diagnostics require dense SVDs, $O(d^3)$ for square matrices; a truncated or randomized approximation must report its rank, residual tolerance and convergence checks, particularly near cutoff ties. The archived study supplies total-delta allocation and maximum-angle drift, but not the full initialization-corrected or chordal report. Those fields require checkpoint evaluation, not imputation from the current ratios.

B.23 ASSETS AND COMPUTE REPORTING

The experiments use public benchmarks, pretrained models, and the PyTorch/Hugging Face ecosystem. The documented hardware includes NVIDIA A100-SXM4-40GB and RTX-3090, but the exported summaries do not assign a verified device to every run. Table 26 exposes the available timing and memory fields. In the inspected runner, time spans the training call, including its validation and checkpoint work; peak memory is CUDA maximum allocated memory measured after training, following a pre-training reset. It is not total process memory, reserved memory, or SVD setup peak. The table therefore supports cost disclosure, not a controlled speed or memory comparison across methods. Token counts and matched LoRA timings are unavailable.

Table 26: Logged training-call minutes and peak allocated GiB (2^{30} bytes) for all 27 archived runs. These are raw resource observations with unverified per-run hardware attribution, not throughput benchmarks. Setup, post-training spectral diagnostics, and total project compute are not reconstructed by these fields.

Task	Condition	Seed	Minutes	Peak allocated GiB
RTE	A	17	12.06	4.408
RTE	A	42	10.44	7.790
RTE	B	17	16.12	4.423
RTE	B	42	11.62	7.787
RTE	C	17	17.46	4.422
RTE	C	42	11.52	7.790
MRPC	A	17	4.96	2.399
MRPC	A	42	2.97	3.737
MRPC	B	17	6.79	2.415
MRPC	B	42	3.89	3.728
MRPC	C	17	6.84	2.418
MRPC	C	42	4.03	3.728
CoLA	A	17	29.22	1.610
CoLA	A	42	32.33	1.599
CoLA	B	17	41.12	1.649
CoLA	B	42	39.62	1.644
CoLA	C	17	42.91	1.650
CoLA	C	42	39.00	1.650
STS-B	A	17	15.85	2.647
STS-B	A	42	18.54	2.637
STS-B	B	17	21.94	2.662
STS-B	B	42	24.14	2.648
STS-B	C	17	22.22	2.663
STS-B	C	42	23.94	2.648
SST-2	A	42	122.96	1.888
SST-2	B	42	164.46	1.922
SST-2	C	42	151.57	1.924

C COMPUTATIONAL COST OF THE SPECTRAL INSTRUMENTS

We distinguish trainable capacity from frozen storage, setup cost, and the computation required to apply the adapter. Let a square layer have width d , selected tail dimension r , and rotated sub-block dimension $q \leq r$. Let b count the token vectors processed by that layer in one forward call. All costs below are per adapted matrix and exclude the base model’s own storage and forward computation, the task head, and hardware-dependent constants.

SVD initialization and frozen storage. A dense SVD of a $d \times d$ matrix has conventional $O(d^3)$ arithmetic cost. For an $m \times n$ matrix with $p = \min(m, n)$, the corresponding economy-SVD cost is $O(mnp)$ and its two bases contain $(m + n)p$ entries. This is a one-time setup cost per distinct weight decomposition, not a decomposition at every optimization step. The practical implementation computes the SVD in float32 before casting stored factors to the adapter dtype, so initialization workspace can exceed persistent basis storage.

A tail-only implementation needs $2dr$ basis entries to apply its update. The implemented leading-plus-tail construction retains the complete bases, totaling $2d^2$ entries, plus $O(d)$ fixed coefficients and any orthogonal-map state. Freezing these entries avoids gradients and optimizer moments for them; it does not remove their memory cost. Setting the leading coefficients to zero in a full-basis implementation does not itself discard those bases or shorten its matrix products. Diagnostics that retain an additional full reference must also be included in memory accounting.

Trainable entries versus effective degrees of freedom. Fixed-tail coefficients require r trainable entries. Two dense $q \times q$ rotation parameters add $2q^2$ stored trainable entries; their orthogonal outputs have only $q(q-1)$ combined manifold degrees of freedom. Ambient input/output scalars add $2d$ entries. Thus the practical unrotated and partially rotated forms store $r+2d$ and $r+2d+2q^2$ trainable entries, respectively, before the task head. A directly trained unrestricted tail core instead has r^2 entries. These counts describe different parameterizations, not equal-capacity interventions. For example, $d = 768$, $r = 256$ gives 1,792 trainable entries per unrotated coordinate-scaled matrix but 1,179,648 frozen basis entries. The GLUE score comparison is not used to infer total training-memory savings.

Forward computation. Applying a diagonal tail update through its thin factors costs $O(bdr)$; ambient scaling adds only $O(bd)$. If partial rotations are materialized into the basis first, the two basis products add $O(dq^2)$ per reconstruction, in addition to the orthogonal-map construction cost. Dense $q \times q$ maps can incur $O(q^3)$ arithmetic. A dense tail core applied between thin basis products instead costs $O(bdr + br^2)$.

The practical forward path applies leading and tail basis products separately; together they span the complete bases. Its adapter branch therefore costs $O(bd^2)$ even when $q = 0$, plus rotation construction when enabled. This is additional to the $O(bd^2)$ base projection. A rank- ℓ LoRA branch costs $O(bd\ell)$, so a smaller trainable count need not mean a cheaper forward pass. Backpropagation also differentiates through the adapter operations, and end-to-end peak memory depends on retained activations, dtype, optimizer state, and checkpointing. The arithmetic counts alone do not establish a measured throughput ratio.

At two bytes per entry, the full frozen bases cost 2.25, 4, and 64 MiB per matrix at widths $d = 768, 1024, 4096$, respectively. Each is twice the storage of the corresponding base projection, and the two basis products add approximately twice its matrix-product arithmetic before rotations or penalties. Across the 48 RoBERTa-base projections, 56.6M basis entries are about 45% of a 125M-parameter backbone’s weight storage at equal dtype; the fresh float32 runs require twice the two-byte figures. No measured usable-width limit is established beyond the tested backbones. At larger widths this quadratic storage and cubic SVD setup favor thin selected-band implementations where the update family permits them, or measuring conventional adapters after training. A strict tail-only branch at $r = d/3$ reduces basis storage by threefold; zeroing a leading core in the existing full-basis path does not realize that saving.

Regularization and diagnostics. Directly forming $M_E = U_L^\top E U_T$ and $M_D = V_L^\top D V_T$ costs $O(dkr)$ for a leading dimension $k = d - r$; evaluating their squared Frobenius penalties then costs $O(kr)$. The fresh runs instead cache the two leading projectors and use the equivalent commutator formula (12). This requires $O(d^2k)$ setup and $O(d^2)$ extra storage per layer, followed by $O(d^2)$ work per step. Raw mixing is measured in every arm; gradients through it are computed only when it is penalized. Neither form requires an SVD of the adapted weight. By contrast, exact ranked-subspace diagnostics use a new dense SVD of $W_0 + \Delta$ and can cost $O(d^3)$ per measured layer/checkpoint. Diagnostic time should be reported separately from optimization time and should not be charged only to the spectral methods in a controlled comparison.

Merging and deployment. At evaluation with dropout disabled, the additive update can be materialized and merged into W_0 . A thin diagonal-tail construction costs $O(d^2r)$ to materialize, whereas the current full-basis product costs $O(d^3)$, plus the rotation and scaler operations; dense materialization requires $O(d^2)$ output storage. After merging and removing the adapter branch, the linear layer has the base model’s shape and inference-time matrix multiplication. This removes the extra branch at deployment, not the preceding SVD, training, or merge costs. In multi-adapter serving, frozen pretrained bases can be shared across tasks; their quadratic storage need not be duplicated per adapter. Keeping branches permits task switching with additional arithmetic, whereas caching separately merged weights requires a dense weight copy per task. The 192-fold branch-arithmetic comparison is not a 192-fold trainable-adapter storage ratio.

D SUPPORTING DOWNSTREAM EVALUATIONS

D.1 GENERATION

Table 27 reports GPT-2 Medium (Radford et al., 2019) on E2E data-to-text generation (Novikova et al., 2017). Fixed-core and partially rotated constructions attain BLEU 68.7 and 68.8, respectively. This extends task-learning evidence beyond classification/regression, without block measurements or a mixing intervention. The imported LoRA and FourierFT entries average three final-epoch runs (Gao et al., 2024); source-reported uncertainties are not a common inferential comparison. The available records do not establish matched training and decoding protocols across these rows, so the table provides historical context rather than a controlled ranking.

Table 27: Supporting GPT-2 Medium E2E results. Non-UI values are from Gao et al. (2024), including their imported full-fine-tuning and adapter results; L and H denote Lin-style and Houlby-style adapters. Uncertainties retain source conventions. Trainable counts exclude frozen-basis storage; these scores do not measure spectral mixing.

Method	Trainable	BLEU	NIST	METEOR	ROUGE-L	CIDEr
Full FT	354.92M	68.2	8.62	46.2	71.0	2.47
Adapter ^L	0.37M	66.3	8.41	45.0	69.8	2.40
Adapter ^L	11.09M	68.9	8.71	46.1	71.3	2.47
Adapter ^H	11.09M	67.3 $\pm$.6	8.5 $\pm$.07	46.0 $\pm$.2	70.7 $\pm$.2	2.44 $\pm$.01
LoRA	0.35M	68.9 $\pm$.3	8.76 $\pm$.06	46.6 $\pm$.1	71.5 $\pm$.1	2.53 $\pm$.03
FourierFT	0.048M	69.1 $\pm$.1	8.82 $\pm$.05	47.0 $\pm$.3	71.8 $\pm$.1	2.51 $\pm$.02
UIOrthoLoRA	0.35M	68.8 $\pm$.1	8.729 $\pm$.06	46.71 $\pm$.3	71.6 $\pm$.02	2.47 $\pm$.01
UILinLoRA	0.078M	68.7 $\pm$.1	8.732 $\pm$.1	46.52 $\pm$.3	71.45 $\pm$.01	2.46 $\pm$.02

E SCOPE OF FUNCTIONAL-RETENTION EVIDENCE

The excluded retention table displayed 40 adapted model–method–capacity–task conditions: 20 rows across Gemma-3-12B-it and Llama-3.2-3B-it, each reporting TriviaQA and HotpotQA. These displayed conditions are not a verified count of training runs. Outcomes had already appeared in an earlier draft, so exclusion was retrospective, not an outcome-blind preregistered decision. The records lack verified sweep selection, matched adaptation and checkpoint-level spectral diagnostics; no reliable full run inventory is available. Displayed outcomes had mixed orderings across settings. The original table remains in the audit archive and contributes to none of this paper’s empirical conclusions. Establishing a functional link requires held-out adaptation and non-trained-example retention on the same geometrically measured checkpoints, with matched tuning and replicated seeds.